The Anatomy of Censorship and Propaganda: Evidence from Russian Wikipedias

Vladimir Avetian[1], Ruben Durante[2], Ulrich Matter[3,5], and Ekaterina Zhuravskaya[4]

[1]University Paris-Dauphine - PSL
[2]National University of Singapore
[3]Bern University of Applied Sciences
[4]Paris School of Economics
[5]University of St.Gallen

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Dictators use censorship and propaganda to shape public opinion. As censored content is unobserved and propaganda can be subtle, uncovering the priorities of dictators is difficult. We address this challenge by comparing two online encyclopedias in Russian: Wikipedia, a crowdsourced platform with independent editors, and Ruwiki, a Russian platform created by cloning Wikipedia and selectively editing content to serve the regime's objectives. Patterns of editing activity suggest that, unlike Wikipedia, Ruwiki is edited by professionals rather than by volunteers. Ruwiki has far fewer contributors who make more edits per person and are active primarily on working days and during working hours. Regarding content, the main differences between the two encyclopedias concern three broad topics: (1) Russia's foreign policy, (2) Russia's domestic politics, and (3) culture. Ruwiki portrays the country as a liberator rather than an invader in both current and past conflicts. It censors pages about the opposition and domestic human rights violations while promoting a benign view of the current political elites. On the cultural dimension, Ruwiki censors articles about LGBTQIA+, sex, and pornography and promotes traditional gender roles, emphasizing the role of Russia as a defender of traditional values in contrast to the perceived Western moral decline.

I Introduction

Authoritarian regimes seek to control the information citizens have access to through censorship (blocking unwelcome content) and propaganda (promoting favorable narratives). Such efforts can permeate all information domains, from news and popular culture to school textbooks and even encyclopedias. A crucial question that remains open is what types of information dictators prioritize when it comes to censorship and propaganda. Answering this question is challenging since censored content is typically hidden from view, and detecting propaganda is difficult due to limited fact-checking opportunities and the high cost of verification.

To shed light on this question, we study the case of competing online encyclopedias – Wikipedia and Ruwiki – in Russia, where the information environment is tightly controlled by the regime. Wikipedia is the world's largest online encyclopedia and consistently ranks among the most popular websites in many countries, including Russia. Wikipedia is free and volunteer-contributed, with independent editors ensuring content accuracy. After several unsuccessful attempts by Russian contributors to alter Wikipedia entries in Russian to fit the regime's views, the entire Russian-language content was cloned in April 2023 to a new platform, Ruwiki.ru. Entries were then selectively deleted and edited. In mid-January 2024 Ruwiki became publicly available (although the editing process has since continued).[1] This case provides an ideal setting to compare potentially manipulated content with a largely unbiased source and understand how information is controlled under an authoritarian regime.

In the first part of our analysis, we compare the organization and functioning of Ruwiki and Russian Wikipedia (henceforth simply Wikipedia). In particular, we examine differences in the content generation process, most notably concerning the number of active contributors and the concentration and timing of edits. Our results indicate several inter-

[1] Although Ruwiki's visitor numbers remain extremely low compared to Wikipedia, it is plausible that the regime could eventually ban Wikipedia from redirecting traffic to Ruwiki including through government-controlled search engines.

esting patterns. First, Ruwiki has a much smaller community of contributors compared to Wikipedia, with fewer contributors making significantly more edits per person. These edits are highly concentrated: the Herfindahl index is 0.14 for Ruwiki versus 0.03 for Wikipedia, and the top 4 contributors account for 46% of edits on Ruwiki compared to 26% on Wikipedia. Furthermore, some of Ruwiki's top contributors exhibit a clear pro-regime slant. The two platforms also differ considerably with regard to the timing of edits. Ruwiki edits are more concentrated on weekdays, with noticeable declines during weekends and holidays, resembling a 9-to-5 work schedule. In contrast, Wikipedia sees more activity during weekends and holidays, as well as evenings, a pattern that is more in line with a volunteer-driven, spare-time effort.

The second part of our analysis investigates what types of information are more likely to be censored or promoted on Ruwiki compared to Wikipedia. To this end, we analyze which entries were deleted or edited on Ruwiki from the time of the initial "fork" to date. We first identify what content features increased the likelihood of an entry being completely removed. Deletions are relatively rare, affecting only around 1,000 articles out of nearly two million on Russian Wikipedia as of the time of the fork. For each entry, we consider the text of the title and of the official categories assigned to it by Wikipedia. We then estimate a series of Lasso regressions to identify which unigrams or bigrams are associated with a higher probability of deletion. The topics most frequently targeted for deletion are: (i) pornography, (ii) LGBTQI+ issues, (iii) Ukraine and the Russian invasion, (iv) political activists, and (v) entries marked as problematic or controversial in the original Wikipedia version.

Finally, we explore which pages underwent edits, either through text deletion or addition. These interventions are far more common than outright deletion, affecting between 50,000 and 100,000 entries. Pages most likely to be edited relate to LGBTQI+ issues, the War in Ukraine, Western sanctions, and the annexation of Crimea and other territories

to Russia. Additionally, entries about certain Russian historical figures, members of the Soviet Union's Communist Party, and Putin's United Russia Party are likely to be edited.

Taken together, our preliminary findings suggest a nuanced, somewhat surprising strategy in Russian propaganda. On the one hand, there is an effort to remove independent information about the Russian invasion of Ukraine and the political opposition, while promoting a favorable image of the Russian political elite. On the other hand, there is a push to promote traditional values in contrast to the perceived moral corruption of the West.

Further analysis will focus on quantifying how propaganda impacts various areas, such as history, philosophy, literature, and religion. Additionally, harnessing the power of LLMs, we will examine the tone of editorial changes to shed light on what narratives and worldviews propaganda seeks to promote or suppress.

II Data and context

Over the past decade, there have been consistent attempts to create clones of Russian Wikipedia. However, only a few projects have progressed beyond the planning stage, such as Runiversalis and Znanie. The landscape shifted dramatically following Russia's invasion of Ukraine, which was extensively covered in Russian Wikipedia. This coverage spurred the development of alternative platforms, with Ruwiki emerging as the most prominent among them. Ruwiki was founded by Vladimir Medeyko, the former head of the Russian chapter of MediaWiki, the non-profit organization that hosts Wikipedia. Medeyko lost his position after announcing the creation of Ruwiki. The foundation of Ruwiki consists of 1.9 million pages forked from Russian Wikipedia on May 1, 2023. Since then, the content has been edited and reviewed by a group of contributors and administrators reportedly controlled by the Russian government. In August 2023, Ruwiki opened its platform for public editing. However, administrative roles remain assigned by

the platform, maintaining centralized control over its operations

A Collection of raw data

To investigate how Ruwiki's editors are modifying the original Wikipedia in Russian knowledge base, we have run a large-scale collection of Ruwiki's (and, prior to the fork, Wikipedia in Russian's) edit logs. To this end, we exploit a technical particularity in how the entire Wikipedia in Russian was copied ("forked") by the initiators behind Ruwiki: Ruwiki has copied not only the webpages containing the Wikipedia in Russian articles at the time of the fork but also the entire technical/data infrastructure behind the official Wikipedia in Russian at the time, including the "backend" with the raw Wikipedia articles in Wikitext format (WikiMedia's markdown format), the entire article revisions histories, and the application programming interface (API) to query the articles' revisions histories. Importantly, this backend infrastructure has not been changed in any relevant way by Ruwiki and was, at least for the duration of our data collection efforts, fully functional and publicly reachable.[2] This allowed us to collect the full raw article revisions–technically the version of an article right after each edit–performed by the editors of Ruwiki.[3] The resulting database thus consists of article versions from the official Wikipedia in Russian, before they were copied to Ruwiki, as well as versions of the same articles after they were copied and altered by Ruwiki editors. In addition, this approach allowed us to systematically extract metadata for each of the edits, such as the timestamp of the edit and the username of the contributor performing the edit. Overall, we have collected over 10.7 million revision entries on 1,729,647 articles.

[2]Note that, unlike the official Wikipedia in Russian (and any other official Wikipedia versions), Ruwiki never offered official download pages for their data. These official download pages (listed under `https://dumps.wikimedia.org/`) are usually the most accessible source of a Wikipedia site's full article and edit logs data and are frequently used in academic research on Wikipedia.

[3]In essence, we use a web crawling procedure that first fetches the unique identifiers of articles stored on Wikipedia in Russian/Ruwiki, and then collects all of the edit logs of each of these articles between the beginning of 2022 and the day of the crawl. Our procedure is highly scalable such that the entire Ruwiki history can be crawled within less than a day. See A.A1 in the Online Appendix for details on the crawling procedure.

B Extraction of relevant articles and detection of article deletions

The collected edit logs also contain observations that are irrelevant to some of our analyses. This involves, for example, articles that are incomplete or ambiguous in format, as well as edit logs that do not contain the full article after the edit. To ensure data quality and relevance, we applied several filters to the crawled raw data to extract our main dataset: articles had to be of relevant length (content longer than 10 characters, i.e., not only consisting of a title), contain at least one heading, and include category tags. The latter is relevant as we build on these categories in some of our analyses.[4] After these filtering steps, we are left with 9.8 million article revision entries on a total of 1,563,203 articles.

These preparatory steps not only allowed us to compile a dataset of relevant article versions before and after the editing by Ruwiki editors, it also provided us with the basis to detect whether an article had been completely removed from Ruwiki after it was forked.

We use a simple procedure to detect articles deleted in Ruwiki. This is possible thanks to the transparency of the original Wikipedia project, which includes a complete dump containing all edits ever made on the platform, along with every version of each article. Specifically, we identify the set of articles that existed in the Russian version of Wikipedia prior to the fork date, May 1, 2023. We then match these articles by their IDs, which are consistently defined for all articles on the official Wikipedia in Russian and on Ruwiki.[5] Our analysis focuses exclusively on nontechnical articles, excluding those that describe Wikipedia's structure rather than the actual content of the encyclopedia. Finally, we define articles as "deleted" if they were present on Russian Wikipedia before the fork but are not present on Ruwiki. To validate this set, we query Ruwiki for each of these articles to double-check their absence. This step helps mitigate false positives that can arise due

[4] The lack of any category fields is also a clear indication that an article is incomplete/faulty.

[5] For exceptional cases where an article was deleted on Ruwiki and later re-created with a new ID — one that does not correspond to the original Wikipedia article ID — we perform additional matching by article names.

to potential instabilities in Ruwiki during initial scraping.

III Analysis of contributors and editing activity

We begin our analysis by examining the activity patterns of contributors to Wikipedia in Russian and Ruwiki over time. Our objective is to determine whether Ruwiki, unlike Wikipedia in Russian, exhibits editing patterns more consistent with a smaller, coordinated group of professional editors rather than a broad-based, volunteer-driven community.

Figure 1 displays the monthly number of active contributors—those who make at least one edit in a given month—on both platforms. The figure shows that Wikipedia in Russian has a large and relatively stable contributor base, commonly exceeding several thousand active users per month. Second, Ruwiki, not surprisingly, starts from a much lower base right after the fork in May 2023 and remains at a consistently smaller scale. This divergence suggests that Ruwiki does not cultivate a wide range of casual volunteer editors (and, that not many switched from Wikipedia in Russian to Ruwiki); rather, its contributor pool appears more restricted and potentially more specialized.

Turning to the intensity of contributions, Figure 2 plots the average number of edits per active contributor monthly. Before May 2023, Wikipedia in Russian displays a fairly stable pattern with contributors averaging on the order of 40–60 edits per month. After the fork, the average activity on Ruwiki surpasses that benchmark, frequently exceeding 70 edits per contributor per month. This contrast underscores how, despite having fewer total contributors, Ruwiki exhibits disproportionately high editing intensity, suggesting that a rather small group of editors is responsible for a lot of of content changes.

Figure 3 further highlights these differences by depicting the distribution of the number of edits per user after May 2023. The top panel (in red) shows Ruwiki's user base, while the bottom panel (in blue) shows that of Wikipedia in Russian. Both histograms

Figure 1. : Active Contributors per Month on Wikipedia in Russian and Ruwiki. This figure displays the number of active contributors per month on Wikipedia in Russian and Ruwiki from January 2014 onwards. An active contributor is defined as a distinct user (by username or IP address) who made at least one edit in a given month. The vertical dashed line marks the initiation of Ruwiki (fork of Wikipedia in Russian) in May 2023. Data for Wikipedia in Russian is represented by open blue circles, while data for Ruwiki (starting from May 2023) is shown with filled red circles. Data were obtained from Ruwiki edit logs (via the Ruwiki API) and the official Wikipedia in Russian dump.

rely on a logarithmic scale for both the number of edits (on the horizontal axis) and the number of users (on the vertical axis). Whereas Wikipedia in Russian has a broad mass of users contributing fewer than 100 edits each, Ruwiki shows a higher concentration of heavy editors (e.g., more than 1,000 edits). This pattern suggests the existence of a small but intense group of contributors on Ruwiki who may be directing content in a more centralized, professionally managed fashion.

A direct way to distinguish between volunteer-driven and potentially professionalized editing is to examine when edits occur. Figure 4 displays the total number of edits aggregated by day of the week for the top 100 contributors (excluding bots) since May 2023. In the upper panel (Ruwiki), editing is distinctly more common on weekdays (especially Wednesday and Friday), while weekend activity (Saturday and Sunday) is comparatively

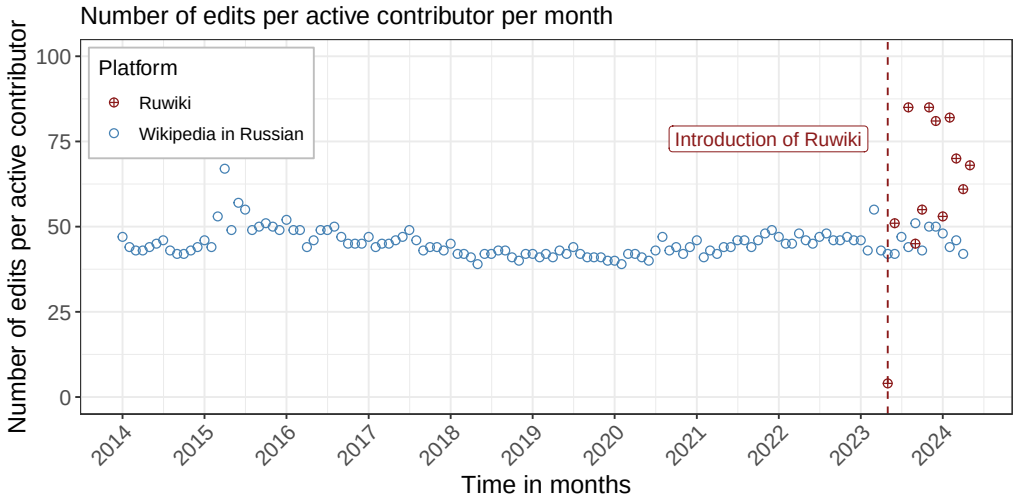


Figure 2. : Average Number of Edits per Active Contributor per Month on Wikipedia in Russian and Ruwiki. This figure displays the average number of edits per active contributor per month on Wikipedia in Russian and Ruwiki from January 2014 onwards. An active contributor is defined as a distinct user (by username or IP address) who made at least one edit in a given month, excluding clearly identified bot accounts (users with 'bot' in their username). The vertical dashed line marks the initiation of Ruwiki (fork of Wikipedia in Russian) in May 2023. Data for Wikipedia in Russian is represented by open blue circles, while data for Ruwiki (starting from May 2023) is shown with filled red circles. Data were obtained from Ruwiki edit logs (via the Ruwiki API) and the official Wikipedia in Russian dump.

low. In the lower panel (Wikipedia in Russian), editing activity is more evenly distributed, with weekend levels rather higher than an average weekday. This contrast aligns with the view that Ruwiki's top editors follow a weekday-dominated 'work schedule," whereas the volunteer base on Wikipedia in Russian continues to produce content across all days.

Finally, to explore editing behavior around an especially prominent holiday period, Figure 5 examines the daily total number of edits around the Russian New Year holidays. The top panel (in red) displays Ruwiki's editing volume, while the bottom panel (in blue) shows Wikipedia in Russian. The vertical dashed lines indicate January 1 (New Year's Day) and January 8 (the day after Orthodox Christmas). As the figure illustrates, Ruwiki experiences a pronounced dip in editing during the holiday intervals, followed by sharp

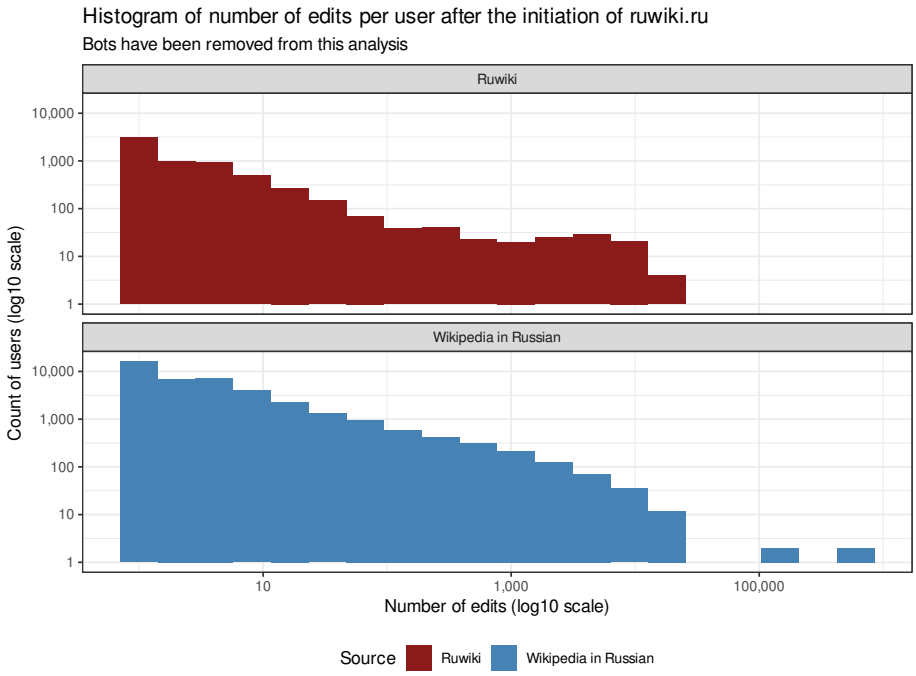


Figure 3. : Distribution of Number of edits per user on Wikipedia in Russian and Ruwiki after 1st May 2023. This figure presents histograms of the number of edits per user on Ruwiki (upper panel, red) Wikipedia in Russian (lower panel, blue) from May 2023 onwards, focusing on pages that are common to both platforms. Edits done by clearly identified bot accounts (users with 'bot' in their username) were excluded from the counts. The horizontal axis displays the number of edits per user on a logarithmic scale, and the horizontal axis shows the count of users on a logarithmic scale. Data were obtained from Ruwiki edit logs (via the Ruwiki API) and the official Wikipedia in Russian dump.

bursts immediately afterward. By contrast, Wikipedia in Russian exhibits a more moderate holiday pattern, with less extreme fluctuations in daily edits. These observations reinforce the notion that editing on Ruwiki follows a more "worklike" rhythm, conforming closely to official holiday schedules. The broader volunteer base of Wikipedia in Russian, on the other hand, appears to spread activity more evenly throughout the holiday period.

Figure 6 underscores the same point by illustrating editing behavior for a 24-hour day. The red bars (left axis) depict Ruwiki, while the blue bars (right axis, scaled by a fac-

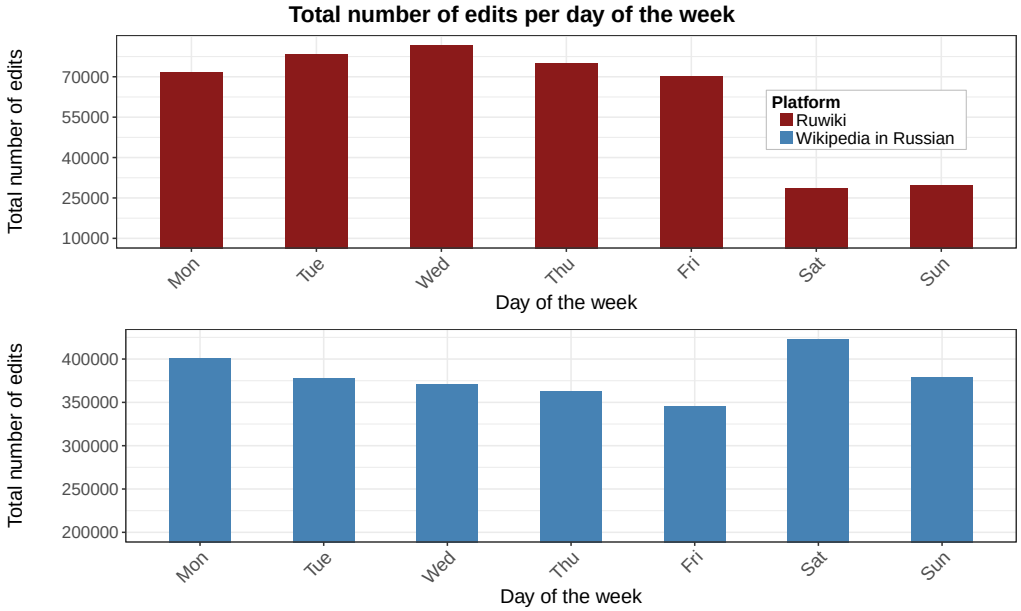


Figure 4. : Total number of edits per day of the week by top contributors on Wikipedia in Russian and Ruwiki. The figure displays the total number of edits per day of the week made by the top 100 contributors (excluding bots) on Wikipedia in Russian and Ruwiki from May 2023 onwards. The top contributors are identified based on their activity on the Russian Wikipedia after the initiation of Ruwiki (fork of Wikipedia in Russian) in May 2023. Edits are aggregated over the period up to the latest common date available in both datasets, with timestamps converted to Moscow time. The bars represent the total number of edits for each day of the week, and the error bars indicate jackknife confidence intervals calculated by systematically excluding each contributor from the analysis. Data for Ruwiki is shown in red in the upper panel and data for Wikipedia in Russian is shown in blue in the lower panel. Data were obtained from Ruwiki edit logs (via the Ruwiki API) and the official Wikipedia in Russian dump.

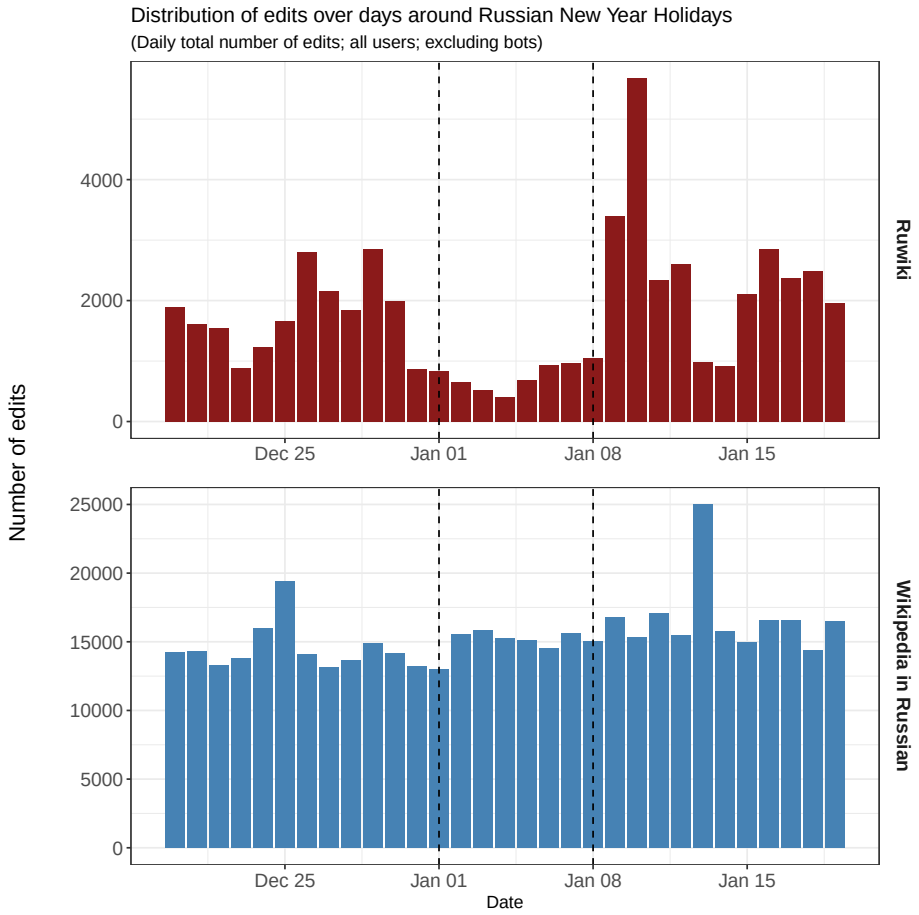


Figure 5. : Daily total number of edits on Ruwiki (red) and Wikipedia in Russian (blue) around the Russian New Year holidays. Vertical lines indicate New Year and Jan 8 (the day after Orthodox Christmas).

tor of 0.2 for clarity) represent Wikipedia in Russian. We observe a pronounced spike in Ruwiki edits during typical daytime working hours (approximately 10:00 to 18:00 Moscow time). Activity diminishes significantly overnight, consistent with professional work schedules. By contrast, Wikipedia in Russian shows a less pronounced drop during the late evening and night hours, suggesting greater participation from non-professional or volunteer contributors, who may edit at varied hours.

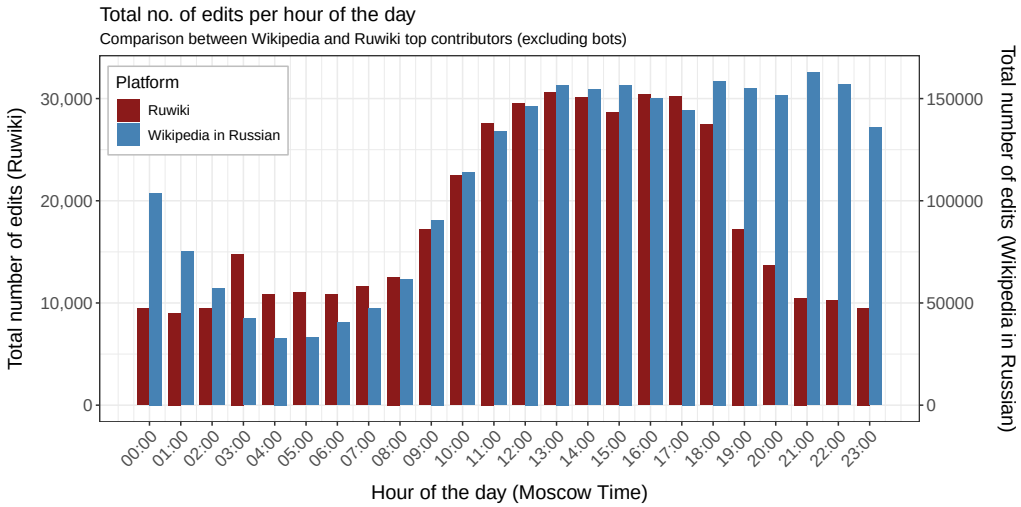


Figure 6. : Total number of edits per hour of the day by top contributors on Ruwiki and Wikipedia in Russian. This figure displays the total number of edits per hour of the day made by the top 100 contributors (excluding bots) on Ruwiki and Wikipedia in Russian from May 2023 onwards. The top contributors are identified based on their activity on the Russian Wikipedia after the initiation of Ruwiki in May 2023. Edits are aggregated up to the latest common date available in both datasets, with timestamps converted to Moscow time. The bars represent the total number of edits for each hour, with data for Ruwiki shown in red and data for Wikipedia in Russian shown in blue. To facilitate comparison between the platforms, the total number of edits for Wikipedia in Russian has been scaled down by a factor of 0.2, and a secondary horizontal axis is provided on the right. Error bars indicate jackknife confidence intervals calculated by systematically excluding each contributor from the analysis. Data were obtained from Ruwiki edit logs (via the Ruwiki API) and the official Wikipedia in Russian dump.

To quantify these descriptive patterns, Figures 7 and 8 present results from regressions

analyzing the effect of day-of-week on (i) the probability of making at least one edit, and (ii) the total size of edits (in bytes). Both specifications include user fixed effects and cluster standard errors at the user level.

Figure 7 shows that the probability of editing is strongly associated with weekdays on Ruwiki, with significant drops on weekends and official Russian holidays (relative to Monday). In contrast, the day-of-week coefficients for the top 100 contributors on Wikipedia in Russian are considerably smaller, indicating less of a weekday-weekend difference. This pattern reinforces the notion that Ruwiki participants follow a more "professional" schedule.

Figure 8 extends the analysis by focusing on the total size of edits per contributor. Again, we see evidence of a professionalized pattern on Ruwiki: total editing volume per user falls appreciably on weekends, while editing on Wikipedia in Russian is comparatively resilient. Taken together, these two sets of regression results confirm that Ruwiki's editing structure aligns more closely with professional or quasi-professional content management.

The patterns across these figures consistently point toward a structured and professionally managed approach to content creation on Ruwiki. While Wikipedia in Russian continues to benefit from the engagement of thousands of distinct contributors, Ruwiki's editing dynamics are dominated by a smaller number of highly active participants, many of whom appear to operate on conventional weekday schedules. These findings align with our broader contention that Ruwiki was set up as a platform to shape and control the narrative under centralized oversight, rather than serving as an open, community-driven alternative to Wikipedia in Russian.

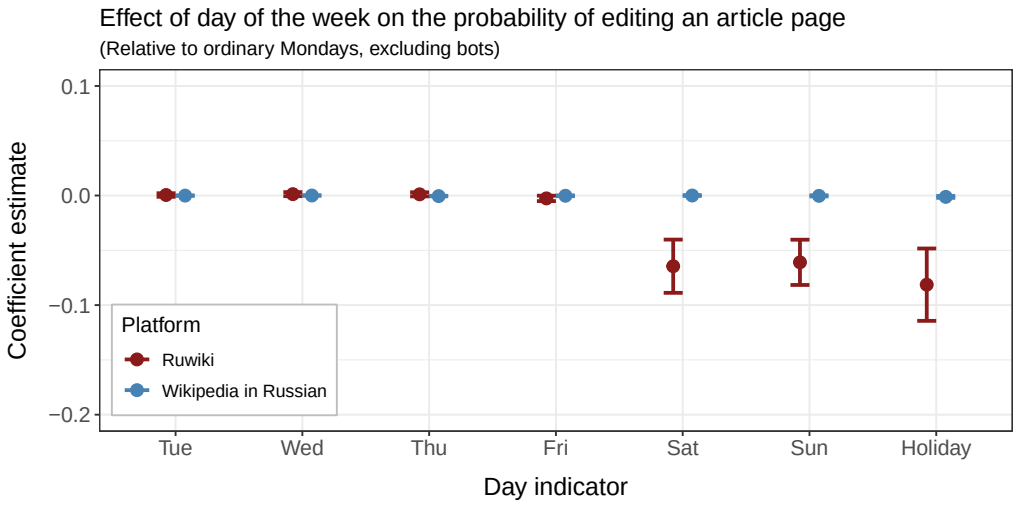


Figure 7. : Effect of day of the week on the probability of editing an article page by top contributors on Ruwiki and Wikipedia in Russian. This figure presents coefficient estimates from regressions analyzing the effect of day of the week on the probability that a contributor edits an article page on Ruwiki and Wikipedia in Russian from May 2023 onwards. The sample includes the top 100 contributors (excluding bots) identified based on their activity on the Russian Wikipedia after the initiation of Ruwiki in May 2023. The dependent variable is a binary indicator of whether a user made at least one edit on a given day. Independent variables are day-type indicators for Tuesday through Sunday and official Russian holidays, with Monday serving as the reference category. The regressions include user fixed effects. The coefficients represent the change in the probability of editing relative to an ordinary Monday. Error bars indicate 95% confidence intervals based on standard errors clustered at the user level. Data are aggregated up to the latest common date available in both datasets, with timestamps converted to Moscow time.

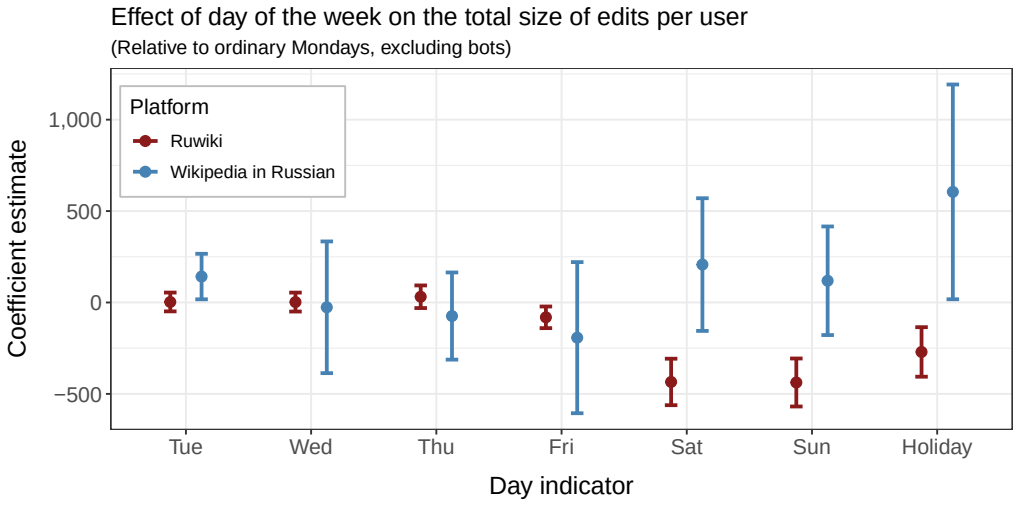


Figure 8. : Effect of day of the week on total size of edits per user on Ruwiki and Wikipedia in Russian. The figure shows coefficient estimates from regressions analyzing the effect of day of the week on the total size of edits per user on Wikipedia in Russian and Ruwiki from May 2023 onwards. The sample includes contributors (excluding bots) who made edits during this period. The dependent variable is the total size of edits (in bytes) made by a user on a given day. Independent variables are day-type indicators for Tuesday through Sunday and official Russian holidays, with Monday serving as the reference category. The regressions include user fixed effects. The coefficients represent the change in the total size of edits per user relative to an ordinary Monday. Error bars indicate 95% confidence intervals based on standard errors clustered at the user level. Data are aggregated up to the latest common date available in both datasets, with timestamps converted to Moscow time.

IV Analysis of Ruwiki article deletions

To identify articles deleted by Ruwiki editors, we leverage the transparency of the original Wikipedia project, which provides a complete record of edits and version of the articles. Using this, we isolate articles that were present on Russian Wikipedia before the fork date but are no longer found on Ruwiki. This process includes matching by article IDs and by names, to account for potential re-creations or renaming. By means of this manual verification, we have identified a total of 2,385 Wikipedia in Russian articles that have been permanently deleted from Ruwiki.

We then manually label each of the articles to get a first descriptive account of what type of article content was targeted. The labels are manually defined by grouping articles with a clear, shared subject that could plausibly explain their deletion. Approximately 6% of articles remain unclassified under this procedure. While this approach relies on the subjective judgment of a human labeler, it provides an illustrative overview of recurring themes among the deleted articles. More objective, data-driven methods are introduced in subsequent sections.Table 1 presents an overview of the resulting categorization.

A Predicting deletion based on content

To better understand what factors drove Ruwiki editors to straight-out delete certain articles, we employ a logistic lasso regression framework aimed at identifying article content that predicts deletion. Building on our list of deleted pages, we prepare a dataset comprising both deleted articles (treatment group) and existing articles (control group). To capture different aspects of the articles's content, we construct four distinct input variants: (i) official Wikipedia in Russian categories, (ii) terms occurring in these categories, (iii) terms occurring in the article title, and (iv) title plus category terms.

As the selection of the control group (articles still existing on Ruwiki) is conceptually not clear, we base our analysis on a bootstrap procedure in which we re-sample a con-

Category	**N**	**%**
Pornography	1,367	57.31
Ukraine	416	17.43
War in Ukraine	376	15.76
-	40	1.67
LGBTQ+	148	6.20
Political regime in Russia	*70*	*2.93*
-	16	0.67
Freedom of speech	14	0.58
Navalny	14	0.58
Human rights	8	0.33
Radical organizations	7	0.29
Propaganda	6	0.25
Opposition	5	0.21
Sexuality	36	1.50
Drugs and addiction	22	0.92
Controversial history	9	0.37
Obscene language	8	0.33
Wikipedia	8	0.33
Chechnya	6	0.25
War crimes	6	0.25
Religion	5	0.21
Feminism	3	0.12
Suicide	2	0.08
Other	*279*	*11.66*
-	149	6.24
Wiki: Proposed deletion	56	2.34
Disambiguation pages	51	2.13
Private company	7	0.29
Wiki: Citing issues	5	0.21
Wiki: Part of larger article	5	0.21
Wiki: Original research	3	0.12
Wiki: Style issues	3	0.12

Total: 2,385 observations

Table 1—: Frequency of subjects among deleted pages, based on a manual labelling process.

trol group of similar size to the treatment group from the population of existing articles.

This approach ensures a balanced dataset for the regression analysis and addresses un-

certainty regarding the appropriate control group. For the term-based input variants, we generate unigrams, bigrams, and trigrams (n-grams) from the preprocessed text and construct a DTM. We remove infrequently occurring terms by setting a sparsity threshold of 99.5%, and exclude common terms that appear in many articles (e.g., ”статьи” [articles], ”списки” [lists], ”год” [year]) to focus on more informative features.

B Regression Model and Bootstrapping

We employ a logistic regression model with an ℓ_1 (lasso) penalty to perform variable selection and estimate the effect of the text features and category names on the probability of article deletion. The model is specified as:

$$\Pr(y_i = 1 \mid \mathbf{x}_i) = \frac{\exp(\beta_0 + \mathbf{x}_i'\beta)}{1 + \exp(\beta_0 + \mathbf{x}_i'\beta)},$$

where y_i is a binary indicator equal to 1 if article i is deleted and 0 otherwise, $\mathbf{x}_i$ is the vector of features for article i, β_0 is the intercept, and β is the vector of coefficients. The feature vector $\mathbf{x}_i$ consists of either binary indicators (for the categories input variant) or term frequencies (for the term-based input variants).[6]

Based on this specification, we run a bootstrap procedure with $B = 100$ iterations. In each iteration, we first randomly sample 5,000 control articles and include all available deleted articles. We then fit the logistic lasso model for each bootstrap sample. This approach allows us to identify features that are consistently predictive of article deletion across different control group samples. From each bootstrap iteration, we keep the nonzero coefficients. Nonzero coefficients correspond to text features (or categories) that are selected as predictive of article deletion. The sign of the coefficient indicates the direction of the effect: In the case of category indicators, a positive coefficient implies that

[6]The lasso penalty term is given by $\lambda\|\beta\|_1$, where λ is the penalty parameter controlling the degree of shrinkage. We use the approach suggested by (Matt Taddy 2017), implemented in the `cv.gamlr` function from the `gamlr` package in R to run the logistic ‘gamma lasso’ regression with cross-validation as a means to select λ. Specifically, we use a 10-fold cross-validation and select the model with the minimum average out-of-sample deviance.

Category Terms		Title and Category Terms	
Term	$\bar{\hat{\beta}}_j$	Term	$\bar{\hat{\beta}}_j$
pornography (порнография)	5.76	**pornography** (порнография)	4.31
sexual (сексуальный)	3.32	**fight** (бой)	3.61
LGBT (лгбт)	2.51	**Populated** (населенный)	3.05
activist (активист)	2.20	**Russian** (российский)	2.66
invasion (вторжение)	1.31	**Ukraine** (украина)	2.66
public (общественный)	1.11	**Russia** (россия)	1.55
caution (осторожность)	1.10	**Point** (пункт)	1.30
Donetsk (донецкий)	1.04	**war** (война)	1.02
Donetsk Region (донецкий область)	1.02	**invasion** (вторжение)	0.50
Russian (российско)	0.99	**invasion Russia** (вторжение россия)	0.44

Table 2—: Top 15 Terms with the largest average coefficients in lasso logistic regressions predicting article deletion. This table reports the top 15 terms with the largest average coefficients from LASSO logistic regression models predicting the deletion of Wikipedia pages. Two input variants are presented: "Category Terms," which utilizes terms extracted from article categories, and "Title and Category Terms," which incorporates terms from both article titles and the article's official Wikipedia in Russian categories. The coefficients represent the average effect of the presence of each term on the log odds of page deletion, calculated over multiple bootstrap samples in which the non-deleted articles were re-sampled from a large set of Wikipedia in Russian articles. Terms are displayed in Russian with English translations in parentheses.

belonging to the category increases the probability of deletion, while a negative coefficient implies the opposite. In the case of term frequency variables, a positive coefficient implies that higher frequencies of the term are associated with an increased probability of deletion, while a negative coefficient suggests that higher frequencies of the term decrease the probability of deletion. The magnitude of the coefficient reflects the strength of the association. At the end of the bootstrap procedure, we collect the coefficients from all iterations and compute the average coefficient values as coefficient estimates and the corresponding standard deviation as the coefficient's bootstrap standard errors.

In addition, we calculate the occurrence percentage of each text feature (or category)—the proportion of bootstrap iterations in which the text feature was selected and only further consider terms that have occurred in 5% of the bootstrap iterations. We then interpret the remaining text features and categories as important predictors of article deletion.

V Analysis of Ruwiki article edits

To analyze the textual changes done by Ruwiki editors, we compare two versions of each article: the latest version just before the fork (the latest version before May 1, 2023), and the latest version after May 1, 2023.[7]

A Computation of article differences (diffs)

For each article, we extract the Wikitext content at both time points. Wikitext is Wikipedia's markup language that includes not only the textual content but also formatting instructions, templates, and other metadata. All data contained in Wikipedia articles is typically stored in this format before it is rendered as web pages. The great advantage of analyzing the data at this level/in this format is that our analyses are not affected by formatting differences in rendered article web pages (HTML files) between the original Wikipedia in Russian and the rather differently structured Ruwiki websites. For both versions of each article, we extract the main text (headers and paragraphs) and remove any Wikitext or HTML formatting syntax.[8]

To identify and highlight the differences between the two versions of each article, we utilize a text comparison algorithm to detect additions and deletions at the word level.[9] The highlighted comparisons are then saved as HTML files. Figure 12 shows examples of such a file based on comparing the main text of the pre-Ruwiki version of the articles on

[7] Our latest collection of all edits on Ruwiki was done on 26th August 2024. Thus, we can consider all edits on Ruwiki that were already stored in Ruwiki's edits backend up until this date.

[8] See Section A.A2 in the Online Appendix for a detailed account of how the raw Wikitext data is processed.

[9] Specifically, we employ the `SequenceMatcher` class in Python's `difflib` library. The `SequenceMatcher` algorithm operates by comparing sequences—such as lists of words or characters—to identify matching blocks and discrepancies. It begins by finding the longest contiguous matching subsequence between the two input sequences. Once this is established, the algorithm recursively applies the same process to the preceding and succeeding segments of the sequences that are not part of the matching block. This method effectively isolates insertions, deletions, and substitutions. In our implementation, we tokenized the texts into lists of words and fed them into the `SequenceMatcher`. The algorithm generated a series of opcodes that describe how to transform the first sequence into the second. Each opcode specifies a particular action—such as `equal`, `delete`, `insert`, or `replace`—along with the indices of the affected elements in both sequences. We used these opcodes to construct an HTML representation of the edited text where deletions from the original text are wrapped in red-highlighted spans and insertions in the new text are wrapped in green-highlighted spans. This visual differentiation allows for an intuitive examination/manual validation of the specific changes made between the two versions of each article.

LGBT, the Katyn Massacre, and the Donbas with the Ruwiki version of the same article after substantive editing by Ruwiki editors. The left panel shows the original file and the right panel a browser-based translation to English.

Importantly, apart from allowing us to identify what textual changes have been made to a particular article, this procedure also helps us filter out articles with edits that did not affect the main text of an article, but other components such as tables, images, references, and links. Overall, we identify 134,060 articles with edits affecting the main text in this way. On average, an article in this set has a main text length of about 700 words (median article length: 315) prior to editing by Ruwiki editors.[10]

Of the 134,060 articles edited by Ruwiki editors, in 63,599 cases the editors deleted part of the main text, in 125,786 of the cases the editors added new text, and in 55,325 cases, there were both additions and deletions.

B Selection of heavily edited articles

Out of the 134,060 articles edited by Ruwiki editors, we select a subset of "heavily edited" articles with either at least 10% of their original text length deleted, at least 10% of their original text length added, or 15% of their original text size either added or deleted on Ruwiki and that have a main text length of over 100 words. The resulting subset consists of 28,118 articles.[11]

Finally, we transform the highlighted textual differences found in these articles into a structured dataset by extracting individual additions and deletions from the HTML diff files. In this dataset, each row consists of an edit for a given Wikipedia article. Text segments that were present in the later version but not in the earlier one are stored in the additions column of this dataset, and text segments that were present in the earlier version but

[10] B3 in the Online Appendix shows the size distribution of these articles before Ruwiki editing in terms of number of characters (panel A) as well as number of words (panel B).

[11] 13,583 of those articles had some deletions by Ruwiki editors, 27,217 of those articles had some additions, and 12,682 had both deletions and additions.

Observance of the rights and freedoms of Russian citizens belonging to the LGBT (homosexual, bisexual and transgender people) is considered in the West a pressing problem in modern Russia. The invisibility and infringement of representatives The propaganda of same-sex relations in Russia is prohibited by law. Most Russian lawyers and human rights activists believe that LGBT representatives in Russia are perceived as a normal phenomenon. This is associated with manifestations of aggression, intolerance and political repression against LGBT in Russia. want special rights for themselves.

(a) LGBT (ЛГБТ)

The Katyn massacre (sometimes also shortened to Katyn - the Katyn crime) was a war crime, mass murder, mass execution of Polish citizens, mainly captured officers of the Polish army, carried out in the spring of 1940 , prisoners of war and civilians, committed in the Katyn forest in the Smolensk region. To this day, the question of who was guilty of this crime remains the subject of scholarly and public debate. According to the first point of view, based on the investigation of the German occupation authorities in 1943. The executions were carried out and according to Soviet documents declassified in 1992 by decision of a special "troika" on the order of Boris Yeltsin, responsibility for the execution is assigned to the NKVD of the USSR in accordance with the resolution of the Politburo of the Central Committee of the All-Union Communist Party (Bolsheviks) of March 5, 1940; executed . According to the second point of view, which was held by the Soviet side of the prosecution at the International Military Tribunal in Nuremberg, the shooting was carried out by Nazi Germany.

(b) Katyn (Катынь)

War on Armed conflict in Donbass , or armed conflict in Donbass (Civil war in Ukraine) - military actions on the territory of Donetsk and Luhansk regions of Ukraine, which began in April 2014 and preceded the full-scale Russian invasion of Ukraine in February military actions in Ukraine since February 2022. It is part of the Russia-Ukraine war . The hostilities were conducted between the Ukrainian armed forces, other security agencies and irregular volunteer formations on one side and pro-Russian armed volunteer formations of the self-proclaimed Donetsk People's Republic (DPR) and Luhansk People's Republic (LPR) . In early 2023, the ECtHR ruled that the territories of eastern Ukraine controlled by the so-called DPR and LPR were under de facto Russian control. The Russian military had been present in these territories since at least April 2014, including a full-scale Russian army presence since August 2014.

(c) Donbas (Донбасс)

Figure 9. : Examples of article diff visualizations for various topics. Each subfigure shows a translated excerpt of detected additions (highlighted in green) and deletions (highlighted in red) in the main text of the articles. The differences are based on comparing the last version of the Wikipedia article in Russian before copying to Ruwiki and the version of the same article on Ruwiki after editing by the Ruwiki editors.

removed in the later one are stored in the deletions column. A replacement/modification of existing content is thus reflected in a row containing both the text removed (in the deletions column) and the text added (in the additions column) as part of the modification.

We then use this dataset to quantify the edits by calculating the number of additions and deletions per article, the total words and characters added or removed, and the overall extent of changes to each article's content. Table 3 presents descriptive statistics of the absolute extent of editing in terms of the number of edits (additions/deletions) and the number of words involved.

	N	Mean	Median	SD	Min	Max
Total no. of edits:	334,201	11.89	4.00	30.48	1	852
... thereof additions:	192,588	6.85	3.00	14.89	0	411
... thereof deletions:	141,613	5.04	0.00	15.96	0	441
Total number of words in edits:	8,888,592	316.12	118.00	779.51	4	22,696
... thereof words added:	5,382,090	191.41	93.00	374.41	0	10,572
... thereof words deleted:	3,506,502	124.71	0.00	584.51	0	22,284

Table 3—: This table presents descriptive statistics of Ruwiki article edits based on a dataset of 28,118 heavily edited articles. The statistics are calculated at the article level, aggregating the edits per article. "Total no. of edits" refers to the total number of individual edits made across all articles. "... thereof additions" and "... thereof deletions" indicate the number of edits that involved adding or deleting text, respectively. "Total number of words in edits" refers to the cumulative number of words affected by the edits, along with their breakdowns into additions and deletions. The statistics provided include the total count (N), mean, median, standard deviation (SD), minimum (Min), and maximum (Max) values across all 28,118 articles.

Table 4 shows statistics of the size of additions and deletions relative to the size of the article's main text previous to the editing by Ruwiki editors (in terms of number of words).

Figure 10 present histograms showing the distribution of words added and deleted as a percentage of initial article text length. The horizontal axes show the percentage of text added/deleted (Panels A and B on a logarithmic (base 10) scale and Panels C and D on a linear scale). The vertical axes indicate the number of cases (frequency) falling into

	Mean	Median	SD	Min	Max
Share words added:	44.41%	30.30%	70.18%	0.00%	4584.35%
Share words deleted:	12.03%	0.00%	22.68%	0.00%	100.00%

Table 4—: This table presents descriptive statistics of Ruwiki article edits based on the dataset of 28,118 heavily edited articles. The statistics are calculated at the article level, aggregating the edits per article. The statistics provided include the mean, median, standard deviation (SD), minimum (Min), and maximum (Max) number words relative to the article's no. of words previous to the changes made by Ruwiki editors.

the corresponding bin (see B4 in the Online Appendix for the histograms based on the fraction of articles instead of the frequency of articles on the vertical scale).

C Predicting text segment additions and deletions based on content

To further understand which type of content is more likely to be censored or promoted on Ruwiki, we analyze the textual content of edits to identify linguistic factors that predict whether a text segment is added or deleted. Based on our collection of text differences (diffs) for heavily edited articles, we construct a dataset at the level of individual diffs with a column containing the raw text contained in the diff, and another column indicating the origin of the diff (the article ID) and an additional column labeling the diff as either an addition or a deletion. To reduce lexical variability and focus on substantive content, we then lemmatize the text segments using the MyStem lemmatizer (Ilya Segalovich 2003), convert all text to lowercase, and remove punctuation, numbers, and stopwords in both Russian and English (Wikipedia in Russian Articles can also contain English phrases). Finally, we generate bigrams and trigrams from the cleaned text.

We employ a Logit Lasso regression to model the probability that a given text segment is an addition or a deletion based on the terms (unigrams, bigrams, or trigrams) occurring in them. The Lasso (Least Absolute Shrinkage and Selection Operator) method introduces an L_1 regularization penalty to the logistic regression, which promotes sparsity

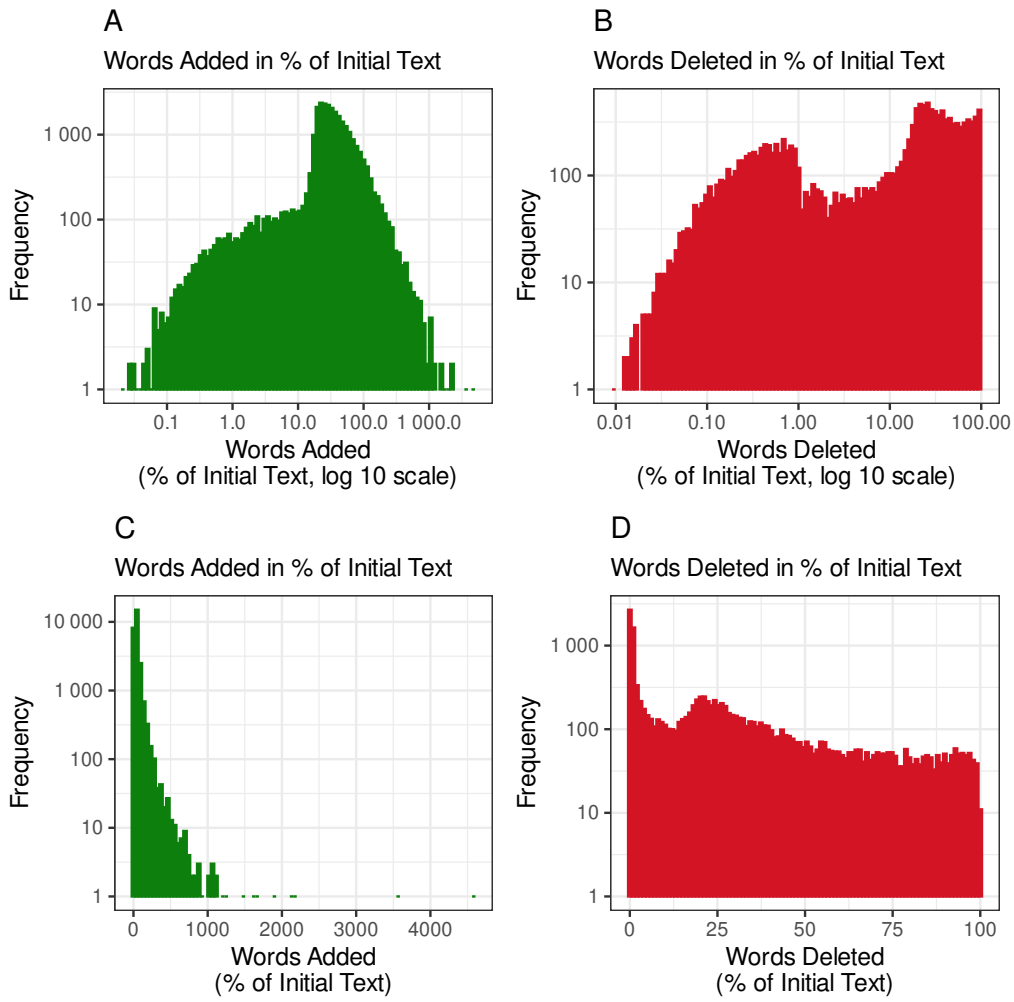


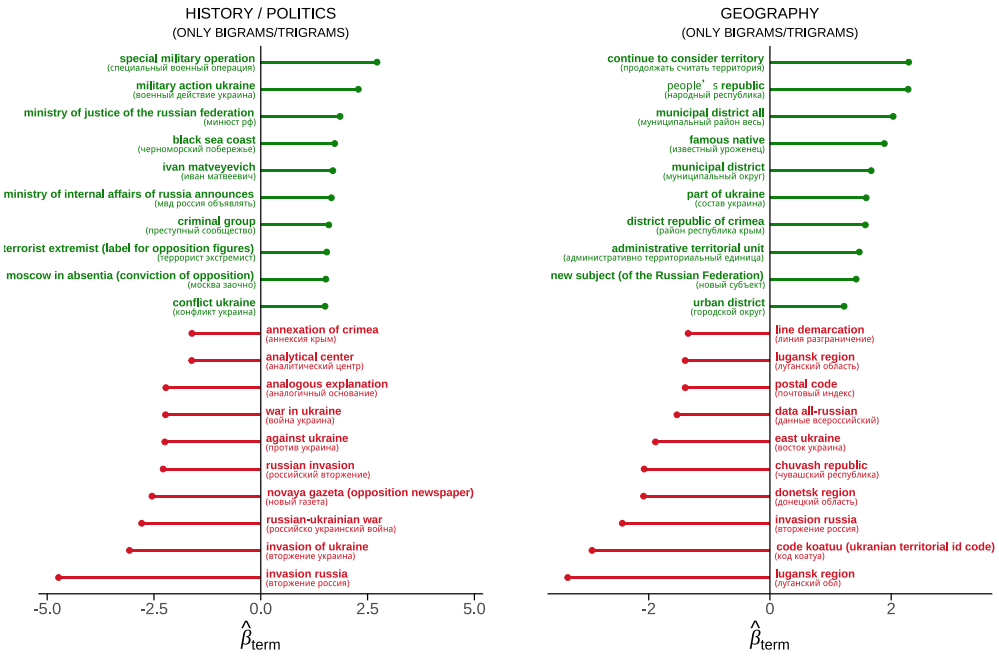
Figure 10. : Distribution of words added and deleted as a percentage of initial article text length. This figure presents histograms of the percentage of words added or deleted relative to the initial number of words in heavily edited articles. Panel A displays the distribution of relative word additions, and Panel B shows the distribution of relative word deletions, both with logarithmic (base 10) scales on the horizontal and vertical axes. Panels C and D present the same distributions with the horizontal axes on a linear scale. Only articles with non-zero word additions or deletions are included in the respective panels.

in the model coefficients. Specifically, the logistic regression model with lasso penalty (formalized as the combined objective function) is:

$$\underset{\beta}{\text{minimize}} \left\{ -\frac{1}{n} \sum_{i=1}^{n} [y_i \log p_i + (1 - y_i) \log(1 - p_i)] + \lambda \sum_{j=1}^{p} |\beta_j| \right\},$$

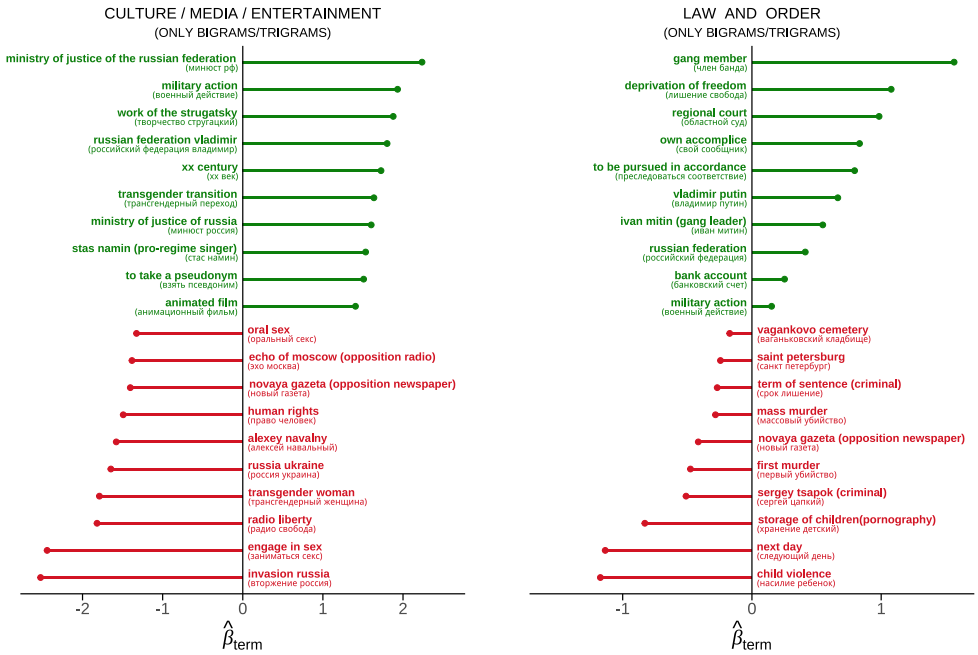
where y_i represents the binary response variable, indicating whether a given text segment is an addition ($y_i = 1$) or a deletion ($y_i = 0$). The predicted probability that the i-th text segment is an addition is denoted by $p_i = \frac{1}{1+e^{-X_i^\top \beta}}$, where X_i is the vector of predictor variables, comprising the lemmatized n-gram features for that observation. The regression coefficients are represented by β, which measure the influence of each n-gram on the probability of a segment being added. The L_1 regularization parameter, λ, controls the strength of the penalty applied to the coefficients, promoting sparsity in the model by shrinking some coefficients to exactly zero. Ten-fold cross-validation is used to select the optimal value of the regularization parameter λ that minimizes the out-of-sample prediction error.

We run the logit lasso regression separately for the different article categories. Figure **??** shows the terms with the ten largest and ten smallest estimated coefficient values for lasso regressions based on bigrams and trigrams in the categories history/politics, culture/media/entertainment/, and geography.



(a) (b)

Figure 11. : Top-10 largest and smallest coefficients for terms predictive of addition or deletion of text segments in articles related to history and/or politics (Panel a) and geography (Panel b). The horizontal bars indicate the estimated logit lasso coefficient values (scale on the horizontal axes). Labels refer to the corresponding term (bigram or trigram) indicators. Terms predictive of text segments being added by Ruwiki editors are shown in green, and terms predictive of text segments being deleted by Ruwiki editors are shown in red.



(a) (b)

Figure 12. : Top-10 largest and smallest coefficients for terms predictive of addition or deletion of text segments in articles related to culture, media, and/or entertainment (Panel a), law and order (Panel b). The horizontal bars indicate the estimated logit lasso coefficient values (scale on the horizontal axes). Labels refer to the corresponding term (bigram or trigram) indicators. Terms predictive of text segments being added by Ruwiki editors are shown in green, and terms predictive of text segments being deleted by Ruwiki editors are shown in red.

VI Ruwiki edits and semantic shifts in historical narratives

To further illustrate how edits by Ruwiki contributors affected content, we turn our attention to the specific domain of history. This focus offers two key advantages. First, history—particularly of the 20th and 21st centuries—is a highly sensitive subject in which the regime has a vested interest in promoting specific narratives. Second, content on Wikipedia and Ruwiki can be systematically compared to a well-defined benchmark: official school textbooks. Indeed, as many commentators have pointed out (BBC Russian, 2023; Deutsche Welle, 2023; Foreign Policy, 2023), the Russian regime has systematically used school textbooks to influence the perception of younger generations, and to advance narratives about Russian and international history that align with its political agenda. Our goal is to test to what extent history-related Wikipedia entries that were heavily edited by Ruwiki contributors became semantically more similar to official school textbooks.

We focus on eight official textbooks aimed at high school students in grades 9 to 11. These books focus on modern and contemporary history (i.e., 19th century to present day), with four dedicated to Russian history, and four to international history. Each textbook is scanned and digitized using OCR technology. Every page of each book is then converted into a numeric embedding[12].

We then apply a similar procedure to the subset of heavily edited entries identified as history-related. For each page, we consider the latest version available on Wikipedia before the fork and the latest version available on Ruwiki after the fork which incorporates all the edits by Ruwiki contributors. Longer entries are split into smaller segments to ensure that their length does not exceed the maximum token limit for embedding generation. The same `text-embedding-3-large` model is then used to create embeddings for

[12]Embeddings are structured, high-dimensional vector representations of text that capture semantic relationships between words and sentences. To generate the embeddings we use OpenAI's `text-embedding-3-large` model, which produces 3,072-dimensional vectors capturing nuanced semantic relationships.

all page segments from both the pre- and the post-edit versions. We use the embeddings space to measure the semantic distance between Wikipedia/Ruwiki entries and textbooks' individual pages (or group of pages).

To measure the distance between Wikipedia/Ruwiki entries and textbooks we use two alternative approaches. First, we measure the similarity between each entry and the textbook page closest to it in the embedding space. This ensures that we compare each entry - both pre- and post-edits - to the page that covers the same (or the most closely related) topic. One limitation of this approach is that the resulting measure can be noisy since the content of a single entry may be covered in multiple textbook pages. To alleviate this concern, we also compare each entry to a group of pages covering the same topic. To this end, we exploit the fact that each textbook is organized in sections and chapters.[13] Specifically, we first identify for each chapter the centroid of all its pages in the embedding space. We then compute, for each entry, the distance to the closest chapter's centroid separately for the the pre-edits version and for the post-edits version. This two procedures provide quantitative measures of how the editorial changes by Ruwiki contributors affect the similarity between each entry and the relevant parts of the textbook.

Figure 13 presents histograms of the Euclidean distance of each entry's embedding and the closest textbook page (left panel) and to the closest chapter's centroid (right panel). The figure overlays the distributions for entry versions before editing (labeled "Wikipedia in Russian") and after editing (labeled "Ruwiki"). Both distributions display substantial overlap, which indicates that the pre- and the post-edits versions are generally similarly close to the textbook.

To gauge how the edits by Ruwiki contributors affect the semantic position of history-related Wikipedia entries relative to textbook, in Table 5 we present the results of a series of regressions. The dependent variable is the normalized Euclidean distance of each

[13] On average each book includes 3.7 sections each composed on average of 6 chapters.

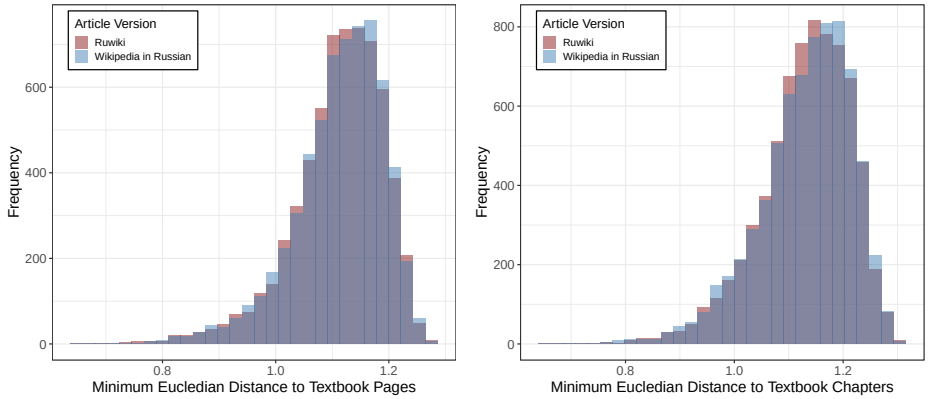


Figure 13. : The figure shows histograms of Euclidean distances between Wikipedia article embeddings and Russian history textbooks in semantic space. The panel on the left shows the minimal distance between each article version and its closest textbook page, the panel on the right shows the minimal distance between each article version and its closest textbook chapter, defined as the centroid of the embeddings of its pages. The "Wikipedia in Russian" distribution (blue) represents the distances before changes by Ruwiki editors, while the "Ruwiki" distribution (red) represents those after edits.

entry version to the closest textbook page (top panel) or to the closest chapter's centroid (bottom). We normalize to ensure that the coefficients reflect changes in units of standard deviations, thus making the interpretation of the magnitude easier.[14] We account for entry-level fixed effects to control for any time-invariant, article-specific characteristics, thus isolating the effect of the editorial intervention on the article's semantic distance to the textbook baseline. Standard errors are clustered at the entry level.

The results in the two panels deliver a similar message. Looking at the full sample (column 1), we see that the coefficient on the post-edit dummy (*Edited by Ruwiki*) is negative and highly significant. This indicate that the editorial changes by Ruwiki contributors made the entries more similar to the content of the textbooks. The following columns report results separetely for different subsamples of entries. We see that the effect is larger for entries that were originally farther away from the textbooks (above-median baseline

[14]We normalize the distance column in our dataset by subtracting the distance mean of the full sample and dividing the difference by the distance standard deviation of the full sample.

distance, column 3) than for those that were closer (column 2). Similarly, it is larger for entries associated solely to the history category (column 4) than for entries associated with multiple categories (column 5). Overall, these findings indicate that edits by Ruwiki contributors made history-related entries more consistent with the official narrative promoted by the regime in school textbooks.

	Dependent variable:					
	Panel A: Minimum Euclidean Distance to Textbook Pages					
	Full Sample (1)	Below Median Dist. (2)	Above Median Dist. (3)	History Only (4)	History & Others (5)	History Only Above Median (6)
Edited by Ruwiki	-0.018^{***}	-0.019^{***}	-0.027^{***}	-0.039^{***}	-0.016^{***}	-0.037^{***}
	(0.004)	(0.006)	(0.004)	(0.011)	(0.004)	(0.012)
Mean Dependent Variable	0	−0.672	0.707	0.089	−0.008	0.654
R^2	0.972	0.937	0.927	0.966	0.972	0.918
Within R^2	0.003	0.002	0.013	0.015	0.002	0.024
Observations	12,514	6,820	6,658	1,082	11,432	618
	Panel B: Minimum Euclidean Distance to Textbook Chapters					
	Full Sample (1)	Below Median Dist. (2)	Above Median Dist. (3)	History Only (4)	History & Others (5)	History Only Above Median (6)
Edited by Ruwiki	-0.010^{***}	0.005	-0.023^{***}	-0.026^{**}	-0.008^{**}	-0.042^{***}
	(0.004)	(0.006)	(0.004)	(0.012)	(0.004)	(0.014)
Mean Dependent Variable	0	-0.681	0.676	0.039	-0.004	0.56
R^2	0.95	0.871	0.881	0.936	0.951	0.832
Within R^2	0	0	0.004	0.003	0	0.008
Observations	19,678	10,818	10,978	1,628	18,050	974

Table 5—: Results from fixed-effects regressions examining the standardized Euclidean distance of Wikipedia articles. In **Panel A**, the dependent variable is the *minimum Euclidean distance between each Wikipedia article's embedding (before or after editing by Ruwiki editors) and the nearest textbook page*. Observations for which the nearest textbook page (before vs. after editing) is not the same are excluded from this panel. In **Panel B**, the dependent variable is the *minimum Euclidean distance between each Wikipedia article's embedding (before or after editing by Ruwiki editors) and the nearest textbook chapter, defined as the centroid of the embeddings of its pages*. Observations for which the nearest textbook chapter (before vs. after editing) is not the same are excluded from this panel. All dependent variables have been normalized, and article fixed effects are included in every model. The *Edited by Ruwiki* coefficient captures the change in semantic proximity following changes by Ruwiki editors, where a negative value indicates movement closer to the history textbook centroid. Due to the normalization, the coefficients reflect the change in the dependent variable in units of standard deviations (of the dependent variable, in the full sample). Different columns represent various subsamples: the full sample, articles split by their initial proximity to the nearest page/centroid (below or above the median distance), articles focusing solely on the history domain, those categorized as history plus other topics, and a subset of history-only articles initially positioned farther from the nearest page/centroid. Standard errors (in parentheses) are robust and clustered at the article level.

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The Anatomy of Censorship and Propaganda: Evidence from Russian Wikipedias

Vladimir Avetian[1], Ruben Durante[2], Ulrich Matter[3,5], and Ekaterina Zhuravskaya[4]

[1]University Paris-Dauphine - PSL

[2]National University of Singapore

[3]Bern University of Applied Sciences

[4]Paris School of Economics

[5]University of St.Gallen

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A Data Appendix

A1 Ruwiki data collection procedure and edit logs data structure

To construct our dataset, we developed an automated crawling procedure running in the cloud. Initially, the procedure retrieves a list of all articles by making requests to Ruwiki's copy of the Wikipedia API, collecting page IDs and titles. Due to API limitations of 500 articles per request, the system iteratively continues this process to encompass the entire article database. The IDs and titles are fed into an AWS SQS (Simple Queue Service) queue.

Once the list is compiled, the system employs AWS Lambda functions—small, event-driven programs that run in the cloud—to process each article's revision history individually (i.e., the Lambda functions run in response to messages received from The AWS SQS queue). The functions fetch the full content and revision history of each article. The collected data, including article content and metadata, is stored in Amazon S3 cloud storage for subsequent filtering and cleaning. To ensure data accuracy and completeness, an additional Lambda function reviews the stored data for any missing or corrupted entries. If issues are detected, the system reprocesses the affected data, maintaining the integrity of the dataset.

*Detailed crawling procedure.*Below we provide a detailed account of how our crawling procedure to collect the article revision histories works:

1) **Initialization and Setup**: For each article identified by its page ID and title, the Lambda function prepares to fetch its revision history starting from a specified date. This date could be the last known revision date from previous data or a fixed start date if we're re-crawling the data. The function sets up the necessary parameters, including the properties to retrieve (such as timestamps, user information, and content) and initializes a session with appropriate headers and proxy settings to comply

with Wikipedia's usage policies.

2) **Constructing API requests**: The function builds requests to the Wikipedia API's `revisions` endpoint. It specifies parameters such as:

   - `action`: Set to "query" to request data.
   - `format`: Specifies the response format, typically "json".
   - `prop`: Set to "revisions" to retrieve revision data.
   - `titles`: The title of the article.
   - `rvprop`: A list of revision properties to fetch (e.g., "ids", "user").
   - `rvlimit`: The maximum number of revisions per request (capped at 50 by the API, or before in case of very large articles).
   - `rvstart`: The timestamp from which to start fetching revisions.
   - `rvdir`: Set to "newer" to retrieve revisions in chronological order.

3) **Fetching revisions in batches**: The function sends the API request and receives a batch of up to 50 revisions. Each batch contains detailed information about the revisions made to the article, including who made the change, when it was made, and the article's content after the revision.

4) **Handling pagination with continuation tokens**: If the article has more than 50 revisions since the start date, the API provides a `continue` parameter in the response. The function uses this token to construct the next request, ensuring it retrieves the next batch of revisions. This loop continues until all revisions are fetched or the most recent revision is reached.

5) **Processing and storing revision data**: After each batch is retrieved:

   - The function extracts the revision data from the response.

- It compiles the data into a structured format, typically a JSON object that includes the page ID, title, and an array of revision details.
- The revision data is then uploaded to Amazon S3, our cloud storage service.

6) **Managing execution time and continuity**: AWS Lambda functions have a maximum execution time limit. The function monitors its remaining execution time, and if it's approaching the limit, it:

   - Updates the metadata in a DynamoDB table with the progress made (e.g., the last revision date fetched).
   - Re-queues a message in the AWS SQS queue with the necessary information to continue fetching revisions from where it left off.
   - Gracefully exits to allow another invocation of the function to pick up the task, ensuring continuity without loss of data.

7) **Error handling and robustness**: Throughout the process, the function includes mechanisms to handle potential errors:

   - **HTTP errors**: Implements retries with exponential backoff for transient network errors like timeouts or server unavailability.
   - **Data validation**: Checks for the presence of expected data fields in the API response to handle any inconsistencies or changes in the API format.
   - **Logging**: Records detailed logs for debugging and monitoring purposes in AWS CloudWatch, aiding in identifying and resolving issues promptly.

8) **Updating metadata**: Upon successfully fetching all revisions:

   - The function updates the DynamoDB metadata table with the total number of revisions collected and the timestamp of the latest revision.

- This metadata is crucial for future data collection cycles, as it allows the system to fetch only new revisions added since the last update, optimizing efficiency.

For each crawling iteration, we have scaled the procedure step-by-step, monitoring server responses and response times to not generate too much burden for the servers running the API. A full crawl of the revisions between, e.g., 2022 and mid-2024 (the relevant time frame for most analyses in the main text) involves about two to three million API requests and amounts to about 100 gigabytes of revisions data.

*Extraction, transforming, and loading of the crawled data data.*After collecting the raw JSON files containing the revision histories of Wikipedia articles from Ruwiki's API, we pre-process the raw data using AWS Glue to transform it into an analytics-friendly format. The transformation involves several key steps:

1) **Data ingestion**: We read all the raw JSON files from our S3 bucket, which contain nested structures of article revisions. Each JSON file includes an array of an article's revision data.

2) **Data partitioning**: To optimize processing performance, we repartition the dataset into 500 partitions. This allows us to leverage parallel processing for the very memory-intense restructuring of the data. Processing one crawling iteration takes about 5-10 minutes when using up to 30 G2X instances (up to 960GB RAM and 240 vCPUs).

3) **Flattening nested structures**: Using a Spark SQL query within AWS Glue, we explode the nested `revisions_data` arrays. This operation transformes each revision within an article into a separate record, effectively flattening the dataset. The query extracts all essential fields from each revision, including:

   - **Article information**:

  - `pageid`: The unique identifier of the article.
  - `title`: The title of the article.

- **Revision details**:
  - `revid`: The unique identifier of the revision.
  - `timestamp`: The timestamp of the revision.
  - `user`: The username of the contributor.
  - `size`: The size of the revision in bytes.

- **Content information**:
  - `contentmodel`: The model of the content (e.g., Wikitext).
  - `content`: The actual content of the revision.

4) **Data type consistency**: To ensure compatibility and ease of analysis, we cast all columns to strings. this step standardized data types across all fields, simplifying downstream processing and querying.

5) **Data cleaning**: We performed cleaning operations on the `pageid` field to address any irregularities. some entries contained extraneous characters such as curly braces, spaces, commas, or the word "null". We applied regular expressions to remove these anomalies, ensuring that `pageid` values were clean and correctly formatted.

The final, transformed dataset was written back to Amazon S3 in Parquet format. Analyses such as counting the number of contributors and measuring the editing activity on Ruwiki are then based on SQL queries against these Parquet files via Amazon Athena.

Parquet is a columnar storage format that offers efficient data compression and allows us to filter the data via queries to AWS Athena.

A2 Processing of Wikitext and detection of textual changes

To analyze the content of the compiled article versions, we proceed in several steps outlined in this subsection.

*Filtering of articles.*First, to ensure data quality and relevance of the articles included in our main content analyses, we apply several filters to the compiled article versions:

- Articles must be of relevant length, with content longer than 10 characters (i.e., not only consisting of a title).
- Articles contain at least one heading, denoted by '==' in the Wikitext.
- Articles include category tags near the end of the content, either in English ('[[Category:') or Russian ('[[Категория:'). This is important, as these categories are used in some of our analyses.

For each qualifying article, we extract two versions: the latest revision before May 1, 2023, and the latest revision after this date.

*Processing of relevant article's Wikitext before and after the fork.*After retrieving the relevant article data, we process the raw Wikitext of an article's version before the fork (last updated version on Wikipedia in Russian) and the latest version after the fork (after editing by Ruwiki editors) to prepare it for textual analysis. This involves converting the Wikitext into clean, standardized HTML format using a locally hosted MediaWiki API (see `https://www.mediawiki.org/wiki/Manual:Installing_MediaWiki` for details). This allows us to parse and convert Wikitext into Wikipedia webpages like in the backend of any Wikipedia version (including Wikipedia in Russian). For most of our textual analyses in the main text, we focus on the article's main text. To do so, we extract the substantive content of each article while removing extraneous elements. The specific processing steps are:

1) **Wikitext conversion**: We convert the raw Wikitext of each article revision into HTML. This conversion is performed by sending the Wikitext to the MediaWiki-API `parse` endpoint, which returns the corresponding HTML representation. This step allows us to maintain consistency with how Wikipedia renders articles, ensuring that the structural elements like headings and paragraphs are appropriately formatted. Importantly, this does not allow us to generate article pages that follow Ruwiki's new design. To avoid problems with consistency, we therefore take a rather conservative approach when it comes to removing layout elements. For our main analyses, we want to avoid considering edits that are due to Ruwiki's different way of formatting stylistic elements and particular page components such as infoboxes but want to focus on changes in the meaning of the article's text.

2) **Content cleaning**: To focus on the main textual content, we implement several pre-processing steps on the Wikitext as well as several cleaning procedures on the HTML output:

   - We strip out templates, infoboxes, navigation boxes, and other embedded objects that are not part of the core article narrative.
   - References to images and files are removed, as these do not contribute to the textual analysis and could introduce noise.
   - All `<ref>` tags and associated content are deleted to avoid including citation details that are not pertinent to the article's main content.
   - We remove category links, metadata, and any remaining HTML comments to prevent non-substantive text from being included.
   - We parse and retain headings (`<h1>` to `<h6>`) and paragraph tags (`<p>`) to preserve the article's structure. however, we remove sections like "References," "External Links," and "See Also" to concentrate on the primary content (we

consider both the English and Russian versions of these keywords as both tend to occur).

3) **Storage**: For each article, we generate two separate HTML files:

   - The version of the latest revision before May 1, 2023 (still on Wikipedia in Russian).
   - The version of the latest revision after May 1, 2023 (maximum up until our last crawl of Ruwiki in August 2024).

*Detection of changes in article texts.*After retrieving and processing the articles into cleaned HTML format, we aim to quantify the textual differences between the versions before and after May 1, 2023 (after editing by Ruwiki editors). To achieve this, we first generate HTML diff files, systematically highlighting the changes. We do so as follows:

1) The script reads the cleaned HTML content of each article from the 'before' and 'after' directories. It performs additional cleaning by removing specific technical phrases and formatting punctuation to enhance comparison accuracy, such as ensuring spaces are appropriately added around punctuation marks.

2) To compute the textual differences between the two versions of each article, we employ Python's `difflib` library, specifically the `SequenceMatcher` class. the `SequenceMatcher` algorithm compares sequences—in our case, lists of words—to identify matching blocks and discrepancies. We tokenize the texts by splitting them into lists of words and feed them into the `SequenceMatcher`. The `SequenceMatcher` works by finding the longest contiguous matching subsequence between the two word lists. once this is established, the algorithm recursively applies the same process to the preceding and succeeding segments of the sequences that are not part of the matching block. This way, the method isolates insertions, deletions, and substitutions.

3) The `SequenceMatcher` generates a series of opcodes that describe how to transform the first sequence into the second. Each opcode specifies a particular action—such as `equal`, `delete`, `insert`, or `replace`—along with the indices of the affected elements in both sequences. This granular information allows us to precisely identify where and how the texts differ.

4) Using the opcodes, we construct an HTML representation of the edited text. Deletions from the original text are wrapped in red-highlighted elements, and insertions in the new text are wrapped in green-highlighted elements. This visual differentiation allows for manual validation of the specific changes made between the two versions of each article while encoding the detected differences in a computer-readable format.

The resulting HTML files serve directly as the basis for several of our textual analyses. For other analyses, we convert the textual differences highlighted in the HTML diff files into a structured dataset, capturing individual additions and deletions at the edit level. The dataset is structured as follows:

- Each observation corresponds to a specific edit made to a Wikipedia article.
- Text segments added in the later version that were absent in the earlier version are recorded in the ”additions” column, while text segments present in the earlier version but removed in the later version are recorded in the ”deletions” column.
- Edits involving content replacement or modification are represented by rows that include both deleted text (in the ”deletions” column) and added text (in the ”additions” column), indicating the changes made to the content.

A3 Data sources and datasets

Our database consists of four main parts:

1) A collection of all Wikipedia in Russian article pages as per 2024-02-01. This data was collected from the official Wikimedia dump: `https://dumps.wikimedia.org/ruwiki/20240201/`. From this data, we compile a table containing all official Wikipedia page categories for each of the Wikipedia in Russian articles at the time. In addition, this data provides a basis to collect data from Ruwiki by providing us with the article IDs as well as article titles (which Ruwiki still used after forking the official Wikipedia in Russian in May 2023).

2) A collection of all Wikipedia in Russian page revisions (covering the entire history of Wikipedia in Russian up until April 2024) collected from the official Wikimedia dump (in April 2024): `https://dumps.wikimedia.org/enwiki/latest/`. This data tells us which Wikipedia contributor edited which (part of) an official Wikipedia in Russian article at which point in time.

3) Based on the data from the official Wikipedia in Russian data (article IDs and article names), we scrape all Ruwiki pages that were part of the fork. The scraping was done in several iterations, starting in mid-March 2024 (about two months after the official Ruwiki "launch"). The main purpose of this extensive data collection was twofold: First, we wanted to make sure that we have a copy of every Ruwiki article page after the launch of Ruwiki (as long as it was also previously present in the official Wikipedia in Russian). Second, the scraping procedure provided a straightforward way to verify whether an article page was deleted after the fork. Either the article page would not be reachable, or we would encounter a placeholder page indicating that there was no such article available.

4) Finally, we exploit the fact that Ruwiki has not only based its main webpage on

Importantly, we run this data collection only for actual article pages. The data dumps of the official Wikipedia in Russian provides data on more than just article pages. In particular, any media file on Wikipedia has, technically, its own webpage. For example, the article on Russia shows a picture of the Russian flag. The file behind this picture has an additional Wikipedia page, simply describing this corresponding image file: `https://ru.wikipedia.org/wiki/ : Flag_of_Russia.svg`.

the fork of Wikipedia in Russian, but it also forked the corresponding Application Programming Interfaces (APIs) provided by Wikimedia. This allowed us to collect the revisions (and, therefore, article histories) of all Wikipedia in Russian articles that were forked by Ruwiki, for the time before and after the fork. This data tells us which Ruwiki editor edited which (part of) an official Wikipedia in Russian article at which point in time. Moreover, we can assess, which parts of an article have been revised after the fork (text added/removed to/from the original Wikipedia in Russian version).

B Additional Figures

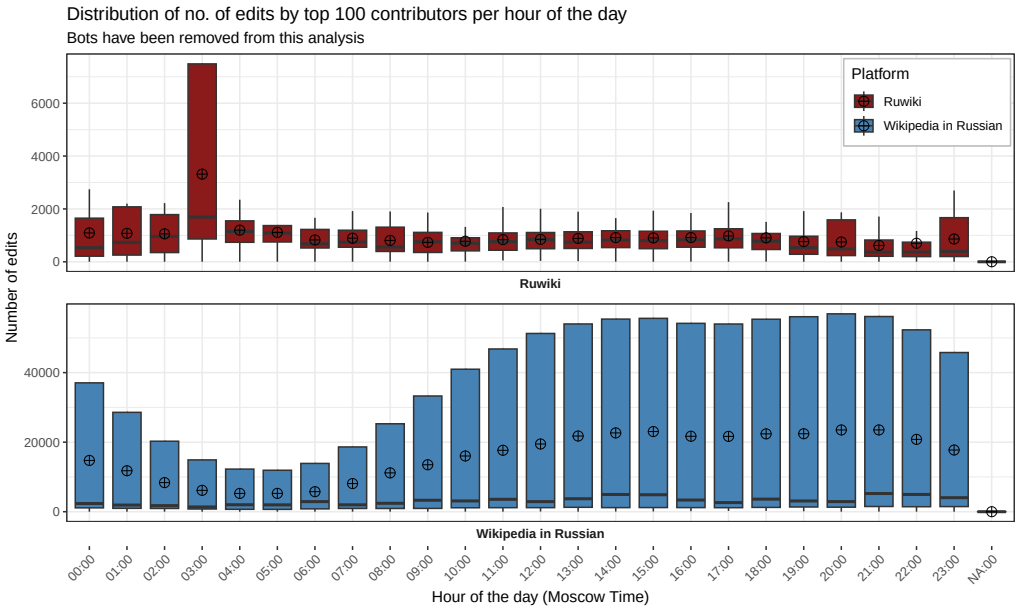

Figure B1. : Boxplots indicating the distribution of editing activity of the top 100 Ruwiki editors (upper panel) as well as the top 100 Wikipedia in Russian contributors by hour of the day. Black dots indicate the mean.

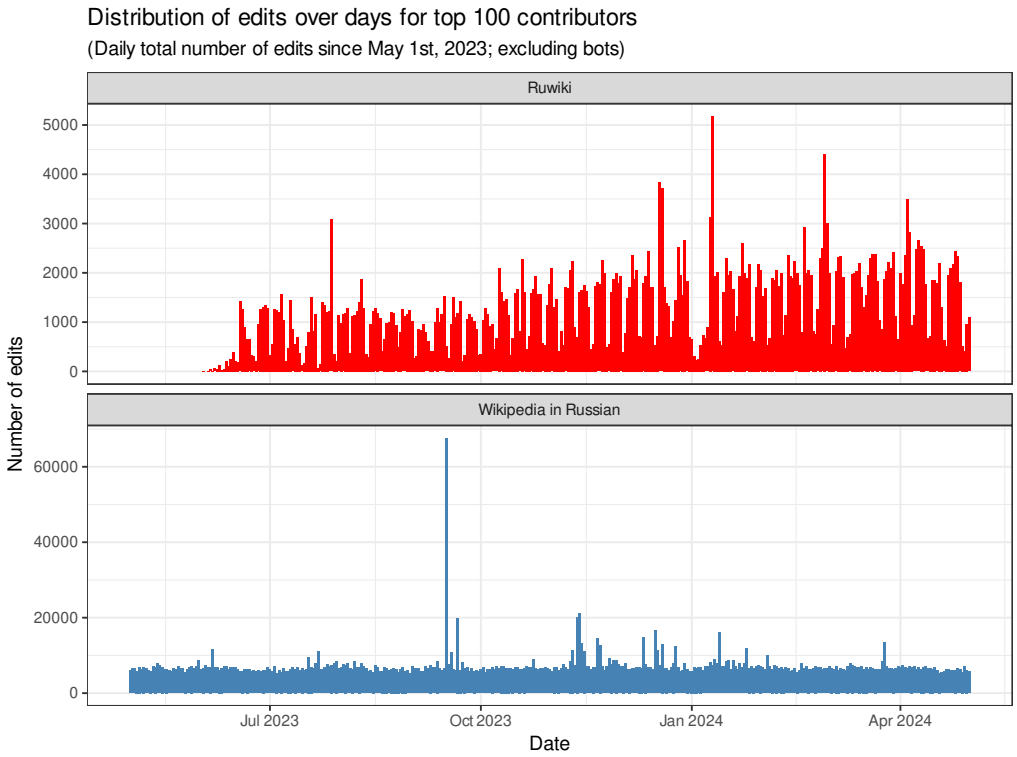

Figure B2. : Daily distribution of edits by top contributors on Ruwiki and Wikipedia in Russian. This figure presents the daily total number of edits made by the top 100 contributors (excluding bots) on Wikipedia in Russian and Ruwiki from May 1, 2023, onwards. The top contributors are identified based on their activity on the Russian Wikipedia after the initiation of Ruwiki in May 2023. Edits are aggregated up to the latest common date available in both datasets, with timestamps converted to Moscow time. Data for Ruwiki is shown in red, data for Wikipedia in Russian is shown in blue (lower panel).

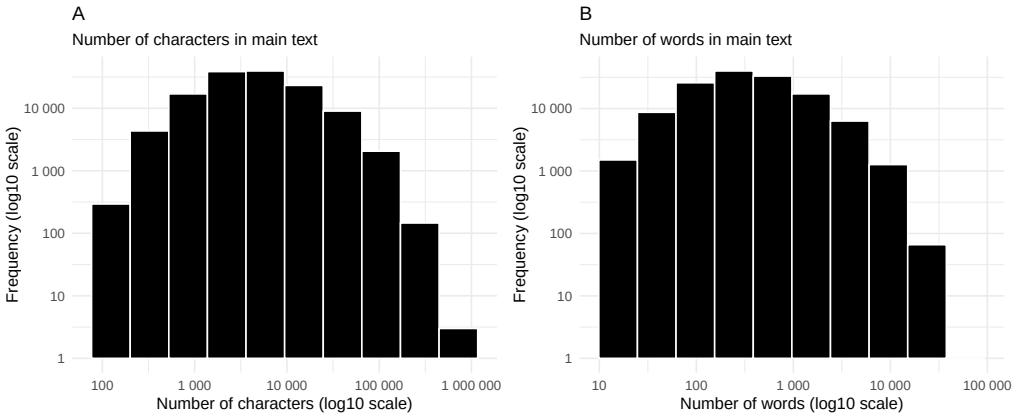

Figure B3. : Distribution of articles' main text lengths in characters and words prior to editing by Ruwiki editors. Panel A shows the histogram of the number of characters in the main text of the 134,060 articles with Ruwiki edits, while Panel B displays the histogram of the number of words. Both axes in the histograms are plotted on logarithmic scales to accommodate the wide range of text lengths.

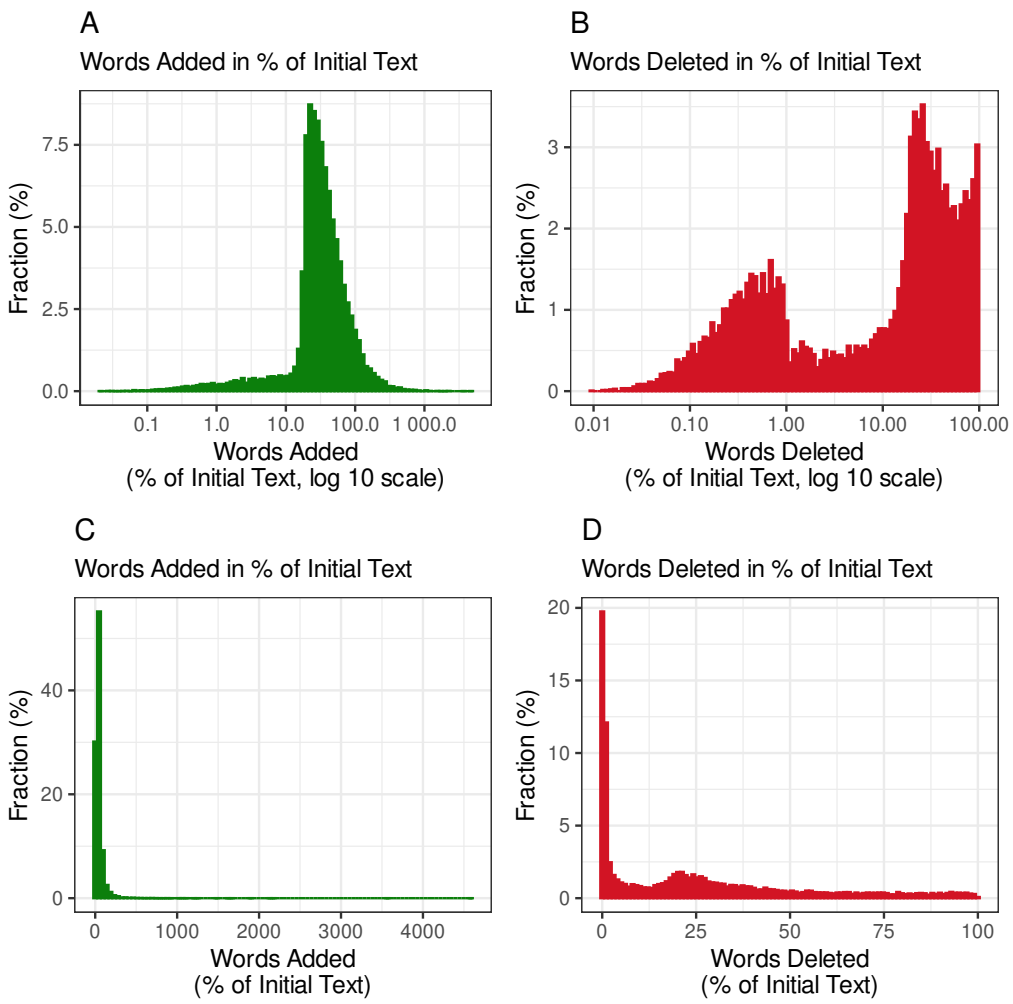

Figure B4. : Distribution of words added (% of overall words added) and deleted as a percentage of initial article text length. This figure presents histograms showing the fraction (in percentage) of heavily edited articles by the percentage of words added or deleted relative to the initial number of characters. Panel A displays the distribution of relative word additions with a logarithmic (base 10) scale on the horizontal axis. Panel B shows the distribution of relative word deletions, also with a logarithmic horizontal axis. Panels C and D present the same distributions with the horizontal axes on a linear scale. The vertical axis in all panels represents the fraction of articles (as a percentage). Only articles with non-zero word additions or deletions are included in the respective panels.

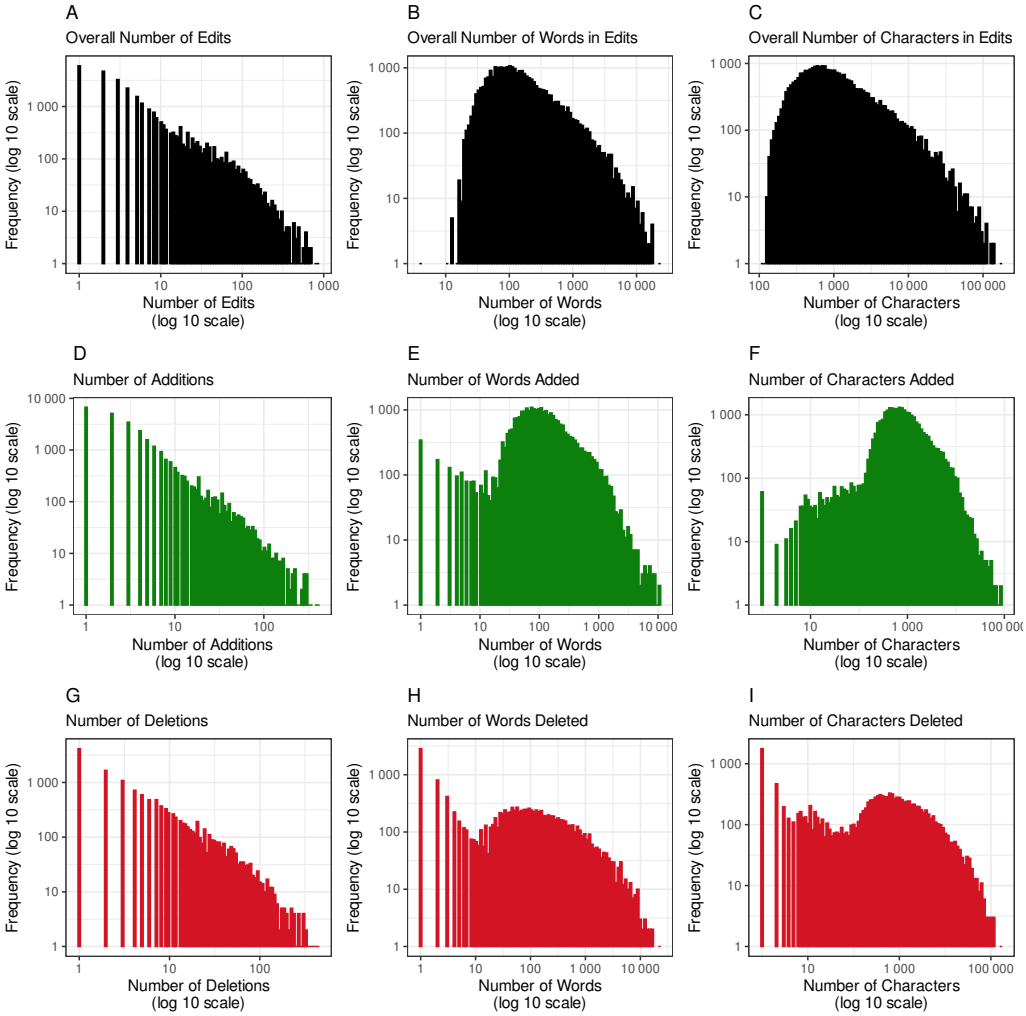

Figure B5. : Histograms illustrating the distribution of edits and textual changes across heavily edited articles. The quantification of edits is based on the main text components of the 28,118 articles identified as heavily edited, ignoring references, tables, info boxes, and links within paragraphs. Horizontal and vertical axes are shown in log-10 scales for ease of readability. Zero values have been removed. Panel (A) shows the distribution of the total number of edits per page, including both additions and deletions. Panel (B) depicts the distribution of the total number of words involved in edits per page. Panel (C) presents the distribution of the total number of characters involved in edits per page. Panels (D)–(F) focus on additions: (D) the number of additions per page, (E) the number of words added per page, and (F) the number of characters added per page. Panels (G)–(I) pertain to deletions: (G) the number of deletions per page, (H) the number of words deleted per page, and (I) the number of characters deleted per page.

C Additional Tables

Rank	Username	Edit Count
1	Мамба41	13023
2	Beginner6767	11005
3	Rwbcontent	10719
4	JoMalone!	10062
5	Александр Мотин	9129
6	Galla	8629
7	TextVoGlave	6883
8	Forum	6435
9	Fox21	5764
10	Forski	5494
11	AnnaG	5370
12	Мирах	5232
13	Sunshine	5137
14	Sabo23	4901
15	Oskk3963	4715
16	Writer	4448
17	LanaKo	4319
18	Perola Barroca	4051
19	Combo19	4036
20	Tester	3910
21	Canopus	3834
22	Ksia	3813
23	VL1738A	3316
24	Rosa	3244
25	Bersene	3072
26	Sharendo	3035
27	DeLo	3033
28	Alena	2904
29	Toxis18	2902
30	Ak	2798
31	Nest leee	2744
32	Blessli	2693
33	Blurs	2549
34	AlAi	2543
35	Horpah	2358
36	Maria444	2270
37	Herbert	2242
38	KORR.RED	2127
39	Veronica	2123
40	Elbevir	2017
41	Ключник	1995
42	DP	1930
43	Red ok	1571
44	Running Rhinos	1517
45	Germiona	1465
46	Anwarsadat	1456
47	Giulini	1425
48	Cherry.mary	1413
49	Unagi	1283
50	TMedvedeva	1253

Table C1—: Top 50 users on Ruwiki (excluding bots, if possible).

	Mean	Median	SD	Min	Max
Characters added:	138.55	25.00	355.36	0	43,457
Characters deleted:	68.71	5.00	496.26	0	72,387
Words added:	19.13	3.00	48.08	0	6,038
Words deleted:	9.42	1.00	67.12	0	10,112

Table C2—: This table presents descriptive statistics of Ruwiki article edits based on a dataset of 134,060 articles identified as having changes to the main text (headers and paragraphs) by Ruwiki editors. The statistics are calculated at the individual edit (either deletion or addition) level.

Predictive of addition	Predictive of deletion
военный действие украина (military action Ukraine)	вторжение россия (invasion Russia)
минюст рф (ministry of justice of the russian federation)	вторжение украина (Invasion of Ukraine)
черноморский побережье (Black Sea coast)	новый газета (new newspaper)
иван матвеевич (Ivan Matveyevich)	российский вторжение (Russian invasion)
мвд россия объявлять (Ministry of Internal Affairs of Russia announces)	война украина (war in Ukraine)
конфликт украина (conflict Ukraine)	против украина (against Ukraine)
преступный сообщество (criminal community)	российско украинский война (Russian-Ukrainian war)
специальный военный операция (special military operation)	аналогичный основание (Analogous basis)
огнестрельный ран (gunshot wound)	аналитический центр (Analytical Center)
террорист экстремист (terrorist extremist)	аннексия крым (annexation of Crimea)
награда звание (award title)	алексей навальный (Alexey Navalny)
штаб квартира (headquarters)	восток украина (East Ukraine)
женатый супруг (married spouse)	движение наш (Our Movement)
военный действие (military action)	действие россия (Action Russia)
набирать голос (gathering votes)	государственный дума (State Duma)
правительство туджман (government Tudjman)	культ личность (Cult of personality)
вооруженный конфликт (armed conflict)	небесный сотня (Heavenly Hundred)
всеобщий выборы (general elections)	чеченский война (Chechen war)
реестр иностранный (foreign register)	республика молдова (Republic of Moldova)
апрель апрель (April April)	эхо москва (Echo Moscow)
новый регион (New Region)	иметь трое (Have three)
специальный военный (special military)	борьба против (Struggle against)
михаил янгель (Mikhail Yangel)	давать момент (to give a moment)
россия ноябрь (Russia November)	гжегожий мотыка (Grzegorz Motyka)
военный учебный (military training)	свой блог (Own blog)
александр голованов (Alexander Golovanov)	право лгбт (LGBT rights)
палата представитель (chamber representative)	донецкий область (Donetsk Region)
вступать должность (to assume office)	согласно расследование (according to the investigation)
украинский сср (Ukrainian SSR)	суверенитет независимость (Sovereignty and independence)
государственный советник (state advisor)	территория украина (Territory of Ukraine)
владимир путин выборы (Vladimir Putin Elections)	ряд сми (a number of media)
отдельный страна (separate country)	максим кац (Maxim Katz)
генрих vi (Henry VI)	честный выборы (Fair elections)
вооруженный сила республика (armed forces of the republic)	российский сми (Russian media)
пост министр (Minister Position)	телеканал дождь (TV channel Dozhd)
диссертация тема (Thesis Topic)	земельный участок (land plot)
сектор газ (gas sector)	гей лесбиянка (gay lesbian)
вынуждать покидать (to force to leave)	территориальный целостность (Territorial integrity)
май президент (May President)	роман шухевич (Roman Shukhevych)
наш эра (Our Era)	левада центр (Levada Center)

Table C3—: **Changes within History and Politics articles:** Top 40 most predictive terms for text additions (left column) and deletions (right column).

Predictive of addition	Predictive of deletion
народный республика (folk republic)	луганский обл (Lugansk region)
продолжать считать (continue to consider)	код коатуа (code KOATUU)
весь входить (all enter)	вторжение россия (invasion Russia)
известный уроженец (famous native)	донецкий область (Donetsk Region)
муниципальный округ (municipal district)	самопровозглашать луганский (to proclaim oneself Luhansk)
состав украина (composition of Ukraine)	чувашский республика (Chuvash Republic)
район республика крым (district Republic of Crimea)	восток украина (East Ukraine)
административно территориальный единица (administrative territorial unit)	ныне непосредственно (now directly)
население численность постоянный (population permanent number)	луганский область (Lugansk region)
новый субъект (new subject)	почтовый индекс (postal code)
городской округ (Urban District)	данные всероссийский (data All-Russian)
фон эппинген (background Eppingen)	поселение упразднение (settlement liquidation)
городской тип (Urban Type)	von eppingen (von Eppingen)
это оспаривать (this is disputed)	находиться контроль (located control)
находиться адрес (located address)	поселковый совет (settlement council)
история основывать (history to establish)	российско украинский (Russian-Ukrainian)
определять территория (define territory)	самопровозглашать донецкий (to proclaim oneself Donetsk)
население согласно (population according)	линия разграничение (line demarcation)
граница сельский (border rural)	состав фаленский (composition of Phalensky)
экономика инфраструктура (economy infrastructure)	сельский совет (Rural Council)
население основной (main population)	местный совет (local council)
военный действие (military action)	являться административный (be administrative)
российский федерация (Russian Federation)	индекс телефонный (index telephone)
херсонский область (Kherson Region)	контроль днр (control DPR)
весь население (entire population)	второй минский (second Minsk)
развитие город (city development)	краснодарский край (Krasnodar Krai)
это место (this is a place)	давать момент (to give a moment)
городской пассажирский (urban passenger)	житель город (resident city)
чело география (people geography)	образование республика (formation republic)
фонд талант (talent fund)	городской совет (City Council)
торговый центр (shopping center)	пункт страна (point country)
мордовия российский (Mordovia Russian)	населять пункт (to populate a point)
первый половина (first half)	село проходить (village pass)
география располагать (geography to locate)	настоящий время (present time)
бурдж халиф (Burj Khalifa)	результат перепись (result census)
влажный континентальный (humid continental)	экономика объект (economy object)
получать название (get a name)	соседний деревня (neighboring village)
общий сложность (overall complexity)	боевой действие (combat action)
начало xx (beginning of the 20th)	герой советский (Hero of the Soviet Union)
самый больший (the biggest)	российский войско (Russian army)

Table C4—: **Changes within Geography articles:** Top 40 most predictive terms for text additions (left column) and deletions (right column).

Predictive of addition	Predictive of deletion
минюст рф (ministry of justice of the russian federation)	вторжение россия (invasion Russia)
военный действие (military action)	заниматься секс (engage in sex)
творчество стругацкий (creativity of the strugatsky)	радио свобода (Radio Liberty)
xx век (XX century)	трансгендерный женщина (transgender woman)
российский федерация владимир (russian federation vladimir)	россия украина (Russia Ukraine)
трансгендерный переход (transgender transition)	алексей навальный (Alexey Navalny)
минюст россия (ministry of justice of russia)	советский российский (Soviet Russian)
стас намин (stas namin)	заявлять свой (to declare one's own)
взять псевдоним (to take a pseudonym)	право человек (human rights)
анимационный фильм (animated film)	новый газета (new newspaper)
игорь верник (igor vernik)	эхо москва (Echo Moscow)
суд москва (court moscow)	гей лесбиянка (gay lesbian)
сериал снимать (shooting a series)	оральный секс (oral sex)
пропаганда лгбт (lgbt propaganda)	сегодняшний день (today)
нетрадиционный сексуальный (non-traditional sexual)	король шут (king fool)
активно писать (actively write)	берлинский кинофестиваль (Berlin Film Festival)
гадкий лебедь (ugly duckling)	майк науменко (Mike Naumenko)
станислав барецкий (stanislav baretsky)	доктор живаго (Doctor Zhivago)
премия просветитель (Enlightener Award)	против украина (against Ukraine)
марго робби (Margot Robbie)	администрация президент (presidential administration)
испанский язык (spanish language)	говорить это (to say this)
дата сентябрь (date september)	российский журналист (Russian journalist)
скончаться октябрь (pass away october)	мужчина женщина (man woman)
работа актер (work actor)	творческий метод (creative method)
скончаться май (pass away may)	исследование проводить (to conduct research)
marvel studios (marvel studios)	издание meduza (publication Meduza)
участие съемка (participation in filming)	российский вторжение (Russian invasion)
отягощать зло (to burden evil)	сцена секс (sex scene)
главный тема (main theme)	комедийный фильм (comedy film)
нарушение российский (violation russian)	группа зоопарк (group zoo)
отправляться рим (to depart for Rome)	игра престол сезон (Game of Thrones season)
де ла (de la)	св себастьян (St. Sebastian)
появляться новый (appear new)	лгбт сообщество (LGBT community)
научный фантастика (science fiction)	сниматься кино (being filmed)
черноморский флот (black sea fleet)	главный редактор (editor-in-chief)
сценарий фильм (script film)	саша белый (Sasha White)
андрей попов (Andrey Popov)	галина юзефович (Galina Yuzefovich)
выход прокат (release in theaters)	субъект российский (subject Russian)
трудный бог (difficult god)	темный лорд (dark lord)
транслироваться канал (broadcast on channel)	открытый гей (open gay)

Table C5—: **Changes within Culture/Media/Entertainment articles:** Top 40 most predictive terms for text additions (left column) and deletions (right column).

Predictive of addition	Predictive of deletion
член банда (Gang member) лишение свобода (Deprivation of Freedom) областной суд (Regional court) становиться известно (become known) преследоваться соответствие (to be pursued in accordance)	насилие ребенок (child violence) следующий день (next day) хранение детский (Storage of children) сергей цапкий (Sergey Tsapkiy) первый убийство (First murder)
владимир путин (Vladimir Putin) иван митин (Ivan Mitin) свой сообщник (Own accomplice) российский федерация (Russian Federation) банковский счет (Bank account)	новый газета (new newspaper) срок лишение (Term of deprivation) массовый убийство (Mass murder) санкт петербург (saint petersburg) ваганьковский кладбище (Vagankovo Cemetery)
авторский право (copyright) александр аверченков (Alexander Averchenkov) борис коровин (Boris Korovin) право смежный (Adjacent right) преследоваться соответствие российский (to be pursued in accordance with Russian)	генеральный прокуратура (Prosecutor General's Office) некоторый данные (Some data) хранение детский порнография (Storage of child pornography) срок лишение свобода (Term of imprisonment)
авторский право смежный (Copyright related) смежный право (Related right) право смежный право (Adjacent rights) российский законодательство (Russian legislation)	

Table C6—: **Changes within Law and Order articles:** Predictive terms for text additions (left column) and deletions (right column).

Predictive of addition	Predictive of deletion
жестовый язык (sign language) удмуртский язык (Udmurt language) исчисление время (Calculation of time) татарский язык (Tatar language) русский население (Russian population)	латинский язык (Latin language) единственный число (Singular number) английский язык (English language) французский язык (French language)
иметь статус (Have status) русский язык (Russian language) смысловой глагол (Semantic verb) ный код (N code) формат исчисление (Format of calculation)	
формат исчисление время (Format of time calculation)	

Table C7—: **Changes within Language articles:** Predictive terms for text additions (left column) and deletions (right column).

Predictive of addition	Predictive of deletion
мочь приводить (can bring) алкогольный напиток (alcoholic beverage) психический расстройство сексуальный (Sexual mental disorder) классификация болезнь (Classification of disease) дефеминизация маскулинизация (Defeminization masculinization)	сексуальный интерес (sexual interest) сексуальный ориентация (sexual orientation) свой сексуальный (his sexual)
фобический расстройство (phobic disorder) маскулинизация мозг (Masculinization of the brain) дефеминизация маскулинизация мозг (Defeminization masculinization of the brain)	

Table C8—: **Changes within Psychology and Psychiatry articles:** Predictive terms for text additions (left column) and deletions (right column).

REFERENCES